

Recall — percentages

A percentage is a fraction with denominator *100* — and everything else follows from that.

LESSON 55

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 10 ENTRY

LESSON CLOCK

10'

12'

12'

6'

What a percentage is	10 min
The three forms	12 min
Finding a percentage of an amount	12 min
One quantity as a percentage of another	6 min

LEARNING OBJECTIVES

- Convert between percentages, fractions and decimals.
- Find a percentage of a quantity.
- Express one quantity as a percentage of another.
- Recognise the percentages worth knowing by heart.

TEXTBOOK

National	pp. 154–157
Cambridge	Stage 7, pp. 96–100
Workbook	Exercise 10.1

01

Explanation

What a percentage is

$$p\% = \frac{p}{100}$$

“Per cent” means “out of a hundred”, so 37% is $\frac{37}{100}$ and nothing more mysterious.

STATEMENT	MEANS
50%	half
25%	a quarter
10%	a tenth
100%	all of it
150%	one and a half times it

PERCENTAGES EXIST TO MAKE COMPARISON EASY

Is $\frac{17}{25}$ better than $\frac{27}{40}$? As 68% and 67.5% the answer is immediate. A common denominator of 100 is the whole point.

The three forms

PERCENTAGE	FRACTION	DECIMAL
10%	$\frac{1}{10}$	0.1
20%	$\frac{1}{5}$	0.2
25%	$\frac{1}{4}$	0.25
50%	$\frac{1}{2}$	0.5
75%	$\frac{3}{4}$	0.75
12.5%	$\frac{1}{8}$	0.125
$33\frac{1}{3}\%$	$\frac{1}{3}$	0.333...

PERCENTAGE TO DECIMAL: DIVIDE BY 100

And decimal to percentage: multiply by 100. The digits move two places, and which way they move is settled by asking whether the answer should be bigger or smaller.

Finding a percentage of an amount

TASK	METHOD	ANSWER
10%	divide by 10	24
20%	double the 10%	48
5%	half the 10%	12
25%	divide by 4	60
15%	10% + 5%	36
37%	$0.37 \cdot 240$	88.8

Every row is a percentage of 240.

BUILD AWKWARD PERCENTAGES FROM EASY ONES

10%, 5% and 1% are all one division away, and 35% is $10\% + 10\% + 10\% + 5\%$. Mental percentage work is nearly all addition.

One quantity as a percentage of another

$$\text{percentage} = \frac{\text{part}}{\text{whole}} \times 100\%$$

PART	WHOLE	WORKING	PERCENTAGE
17	25	$\frac{17}{25} \cdot 100$	68%
27	40	$\frac{27}{40} \cdot 100$	67.5%
18	24	$\frac{18}{24} \cdot 100$	75%
9	60	$\frac{9}{60} \cdot 100$	15%

THE WHOLE GOES UNDERNEATH

“What percentage of 40 is 27” puts 40 in the denominator. Swapping them gives 148%, which the sense check rejects at once.

02 Worked examples**EXAMPLE 1** Find 15% of 240.

① 10% is 24.

② 5% is half of that, 12.

③ $24 + 12 = 36$

ANSWER

36

EXAMPLE 2 What percentage of 40 is 27?

① $\frac{27}{40}$

The whole underneath.

② $\cdot 100 = 67.5$

③ 67.5%

ANSWER

67.5%

EXAMPLE 3 Which is better, 17 out of 25 or 27 out of 40?

① $\frac{17}{25} = 68\%$

② $\frac{27}{40} = 67.5\%$

- ③ The first, by half a percentage point.
Impossible to see without converting.

ANSWER

17 out of 25

03 Interactive model

Put two test scores out of different totals on the board and ask which is better; the class reaches for percentages without being told to.

Percentage, fraction, decimal

25% as a fraction:

$$\frac{1}{4}$$

$$\frac{1}{25}$$

$$\frac{25}{1}$$

$$\frac{1}{5}$$

0.2 as a percentage:

2%

20%

0.2%

200%

10% of 240:

2.4

24

240

12

5% of 240:

6

12

24

48

15% of 240:

30

36

40

48

27 as a percentage of 40:

67.5%

148%

54%

75%

18 as a percentage of 24:

60%

75%

80%

133%

150% of a number is:

half of it

one and a half times it

impossible

the same

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Per cent	Yuzdan	Со ста
Fraction	Kasr	Дробь
Decimal	O'nli kasr	Десятичная дробь
To convert	O'girmoq	Преобразовать
Of a quantity	Miqdordan	От величины
Whole	Butun	Целое
Equivalent	Teng kuchli	Равный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

37% means:

$$\frac{37}{10}$$

$$\frac{37}{100}$$

$$\frac{100}{37}$$

37

0.75 as a percentage:

7.5%

75%

0.75%

750%

1 8 as a percentage:

8%

12.5%

18%

1.8%

25% of 240:

48

60

80

96

To write a part as a percentage you divide by:

the part

the whole

100

the difference

9 as a percentage of 60:

9%

15%

54%

6.7%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 25% as a fraction

ANSWER $\frac{1}{4}$

2. 20% as a decimal

ANSWER 0.2

3. 0.75 as a percentage

ANSWER 75%

4. $\frac{1}{8}$ as a percentage

ANSWER 12.5%

5. 10% of 240

ANSWER 24

6. 25% of 240

ANSWER 60

7. 50% of 240

ANSWER 120

Medium · the standard question

7 PROBLEMS

1. 5% of 240

ANSWER 12

2. 15% of 240

ANSWER 36

3. 37% of 240

ANSWER 88.8

4. 17 as a percentage of 25

ANSWER 68%

5. 27 as a percentage of 40

ANSWER 67.5%

6. 18 as a percentage of 24

ANSWER 75%

7. 9 as a percentage of 60

ANSWER 15%

Hard · stretch and reason

7 PROBLEMS

1. 35% of 240 by building from 10%

ANSWER 84

2. 12.5% of 320

ANSWER 40

3. Which is better, $17/25$ or $27/40$?

ANSWER $17/25$, at 68%

4. 150% of 60

ANSWER 90

5. 1% of 240, and hence 3%

ANSWER 2.4 and 7.2

6. A test mark of 45 out of 60 as a percentage

ANSWER 75%

7. Why do percentages make comparison easy?

ANSWER Every fraction is put over the same denominator, 100

07 Homework

Homework — 5 tasks

Build awkward percentages from 10%, 5% and 1%.

1. Convert 45%, 8% and 130% to decimals and to fractions in lowest terms.
2. Find 10%, 5% and 15% of 360.
3. Find 35% of 480.
4. Write 21 as a percentage of 35.
5. Which is the better mark: 19 out of 25 or 30 out of 40?

Multiplying and dividing by powers of ten

A Cambridge insert: the digits move, the point stays — and why that is the better picture.

LESSON 56–57

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 3.1

LESSON CLOCK

54'

21'

5'

Place value	54 min
Converting units	21 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply and divide by *10*, *100* and *1000*.
- Multiply and divide by *0.1* and *0.01*.
- Explain the effect in terms of place value.
- Use the rules to convert between units.

TEXTBOOK

National	pp. 158–163
Cambridge	Stage 7, pp. 32–36
Workbook	Exercise 3.1

01

Explanation

Place value

COLUMN	HUNDREDS	TENS	UNITS	TENTHS	HUNDREDTHS
value	100	10	1	$\frac{1}{10}$	$\frac{1}{100}$
3.47	—	—	3	4	7
347	3	4	7	—	—

EACH COLUMN IS TEN TIMES THE ONE ON ITS RIGHT

Multiplying by *10* moves every digit one column to the left, which is the same thing as moving the point one place to the right. The first description is the true one; the second is a shortcut.

Multiplying and dividing by 10, 100, 1000

START	$\times 10$	$\times 100$	$\div 10$	$\div 100$
3.47	34.7	347	0.347	0.0347
58	580	5800	5.8	0.58
0.6	6	60	0.06	0.006

ZEROS HOLD THE EMPTY COLUMNS

$0.347 \div 10$ is 0.0347 : a zero has to appear in the tenths column, because that column is now empty. Leaving it out changes the number tenfold.

By 0.1 and 0.01

Multiplying by 0.1 is the same as dividing by 10 — and dividing by 0.1 is the same as multiplying by 10 .

OPERATION	SAME AS	EXAMPLE
$\times 0.1$	$\div 10$	$47 \cdot 0.1 = 4.7$
$\times 0.01$	$\div 100$	$47 \cdot 0.01 = 0.47$
$\div 0.1$	$\times 10$	$47 \div 0.1 = 470$
$\div 0.01$	$\times 100$	$47 \div 0.01 = 4700$

MULTIPLYING CAN MAKE A NUMBER SMALLER

$47 \cdot 0.1 = 4.7$. “Multiplication makes bigger” is only true for multipliers above 1 — one of the most useful facts of the year to unlearn.

Converting units

CONVERSION	OPERATION	EXAMPLE
m to cm	$\times 100$	$3.5\text{ m} = 350\text{ cm}$
cm to m	$\div 100$	$47\text{ cm} = 0.47\text{ m}$
km to m	$\times 1000$	$2.4\text{ km} = 2400\text{ m}$
g to kg	$\div 1000$	$750\text{ g} = 0.75\text{ kg}$
l to ml	$\times 1000$	$1.5\text{ l} = 1500\text{ ml}$

SMALLER UNIT, BIGGER NUMBER

A metre is many centimetres, so the number of centimetres is larger. Deciding whether to multiply or divide is a question about the units, not about the digits.

02 Worked examples

EXAMPLE 1 Compute $3.47 \cdot 100$ and $3.47 \div 100$.

- 1 Each digit moves two columns left: 347.
- 2 Each digit moves two columns right: 0.0347.
Zeros fill the empty columns.

ANSWER
347 and 0.0347

EXAMPLE 2 Compute $47 \div 0.1$.

- ① Dividing by 0.1 is multiplying by 10 .
- ② How many tenths are in 47 ? Ten in each unit.
- ③ 470

ANSWER 470 **EXAMPLE 3** Convert 750 g to kilograms and 2.4 km to metres.

- ① A kilogram is 1000 g, so divide: 0.75 kg.
- ② A kilometre is 1000 m, so multiply: 2400 m.

ANSWER 0.75 kg and 2400 m**03** Interactive model

Write the place-value columns once on the board and move the digits along it rather than the point; the zeros then appear where they are needed without a rule.

Which way do the digits move?

$3.47 \cdot 10$ equals:

34.7

0.347

347

3.470

$3.47 \div 100$ equals:

0.347

0.0347

347

34.7

$58 \div 10$ equals:

5.8

0.58

580

5800

$47 \cdot 0.1$ equals:

470

4.7

0.47

4700

$47 \div 0.1$ equals:

4.7

470

0.47

4700

$47 \div 0.01$ equals:

470

4700

0.47

4.7

3.5 m in centimetres:

35

350

0.035

3500

750 g in kilograms:

7.5

0.75

75

0.075

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Place value	Xona qiymati	Разрядное значение
Power of ten	O'nning darajasi	Степень десяти
Decimal point	O'nli nuqta (vergul)	Десятичная запятая
To shift	Siljimoq	Сдвигаться
Tenth	O'ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Zero as a placeholder	O'rin to'ldiruvchi nol	Ноль как разделитель
Unit conversion	Birliklarni almashtirish	Перевод единиц

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Multiplying by 10 moves each digit:

one column left

one column right

two left

nowhere

Dividing by 100 moves each digit:

two left

two right

one right

nowhere

Multiplying by 0.1 is the same as:

dividing by 10

multiplying by 10

adding 0.1

nothing

Dividing by 0.01 is the same as:

dividing by 100

multiplying by 100

multiplying by 0.01

nothing

$0.347 \div 10$ equals:

0.0347

0.347

3.47

0.00347

To change metres to centimetres:

divide by 100

multiply by 100

multiply by 10

divide by 10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3.47 \cdot 10$

ANSWER 34.7

2. $3.47 \cdot 100$

ANSWER 347

3. $3.47 \div 10$

ANSWER 0.347

4. $3.47 \div 100$

ANSWER 0.0347

5. $58 \cdot 100$

ANSWER 5800

6. $58 \div 10$

ANSWER 5.8

7. $0.6 \cdot 100$

ANSWER 60

Medium · the standard question

7 PROBLEMS

1. $47 \cdot 0.1$

ANSWER 4.7

2. $47 \cdot 0.01$

ANSWER 0.47

3. $47 \div 0.1$

ANSWER 470

4. $47 \div 0.01$

ANSWER 4700

5. 3.5 m in cm

ANSWER 350

6. 47 cm in m

ANSWER 0.47

7. 750 g in kg

ANSWER 0.75

Hard · stretch and reason

7 PROBLEMS

1. $0.06 \div 0.1$

ANSWER 0.6

2. $0.06 \cdot 1000$

ANSWER 60

3. 2.4 km in cm

ANSWER 240 000

4. 1.5 l in ml

ANSWER 1500

5. $0.008 \cdot 100$

ANSWER 0.8

6. Why does $47 \cdot 0.1$ make the number smaller?

ANSWER The multiplier is less than one

7. 3 200 ml in litres

ANSWER 3.2

07 Homework

Homework — 5 tasks

Ask first whether the answer should be bigger or smaller; that settles the direction.

1. Compute $5.28 \cdot 10$, $5.28 \cdot 1000$ and $5.28 \div 100$.
2. Compute $36 \cdot 0.1$ and $36 \div 0.01$.
3. Convert 4.7 m to centimetres and 86 cm to metres.
4. Convert 3.6 kg to grams and 450 ml to litres.
5. Explain in one sentence why $\div 0.1$ makes a number larger.

Rounding to a given accuracy

A Cambridge insert: which digit decides, and what a rounded answer really claims.

LESSON 58

1 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 3.2

LESSON CLOCK

10'

12'

12'

6'

The deciding digit	10 min
Decimal places	12 min
Significant figures	12 min
Estimating	6 min

LEARNING OBJECTIVES

- Round to the nearest ten, hundred and thousand.
- Round to a given number of decimal places.
- Round to a given number of significant figures.
- Estimate a calculation by rounding first.

TEXTBOOK

National	pp. 164–166
Cambridge	Stage 7, pp. 36–40
Workbook	Exercise 3.2

01

Explanation

The deciding digit

Look at the digit **immediately after** the place you are rounding to. If it is 5 or more, round up; otherwise leave the digit alone.

NUMBER	TO THE NEAREST TEN	HUNDRED	THOUSAND
3 472	3 470	3 500	3 000
2 851	2 850	2 900	3 000
795	800	800	1 000

ONLY ONE DIGIT DECIDES

For 3 472 to the nearest hundred, the deciding digit is the 7, not the 2. Looking further along and “rounding twice” gives the wrong answer.

Decimal places

NUMBER	1 D.P.	2 D.P.	WHOLE NUMBER
3.472	3.5	3.47	3
0.0851	0.1	0.09	0
9.96	10.0	9.96	10
2.345	2.3	2.35	2

9.96 TO ONE DECIMAL PLACE IS 10.0, NOT 10

The rounding can carry into the next column, and the zero must stay to show the accuracy claimed. Writing 10 claims a whole number; 10.0 claims a tenth.

Significant figures

Significant figures are counted from the first non-zero digit, whatever the size of the number.

NUMBER	1 S.F.	2 S.F.	3 S.F.
3 472	3 000	3 500	3 470
0.0851	0.09	0.085	0.0851
205.6	200	210	206

LEADING ZEROS ARE NOT SIGNIFICANT; ZEROS BETWEEN DIGITS ARE

In 0.0851 the first significant figure is the 8. In 205.6 the 0 counts, because it sits between two significant digits.

Estimating

CALCULATION	ROUNDED	ESTIMATE	EXACT
$38 \cdot 21$	$40 \cdot 20$	800	798
$4\,187 \div 21$	$4\,000 \div 20$	200	199.4
$6.9 \cdot 5.2$	$7 \cdot 5$	35	35.88
$297 + 405$	$300 + 400$	700	702

AN ESTIMATE CATCHES THE BIG MISTAKES

It will not tell you whether the answer is 798 or 799, but it tells you at once that 7 980 is wrong. Estimating first is how a calculator error is caught.

02 Worked examples

EXAMPLE 1 Round 3 472 to the nearest hundred and to 2 significant figures.

- 1

Nearest hundred: the deciding digit is 7, so round up.
3 500
- 2

Two significant figures: the first two digits are 3 and 4; the next is 7.
3 500

ANSWER

3 500 both times

EXAMPLE 2 Round 9.96 to one decimal place.

- 1 The deciding digit is the 6.
- 2 Round the 9 up: it becomes 10 and carries.
- 3 10.0
The zero shows the accuracy claimed.

ANSWER

10.0

EXAMPLE 3 Estimate $4\,187 \div 21$.

- 1 Round to one significant figure each: $4\,000 \div 20$.
- 2 = 200
- 3 The exact value is 199.4 — close enough to check by.

ANSWER

About 200

03 Interactive model

Ask for an estimate before every calculation for a week; the habit catches more errors than any amount of rechecking.

Which digit decides?

3 472 to the nearest hundred:

3 400

3 500

3 470

3 000

2 851 to the nearest thousand:

2 000

3 000

2 900

2 800

3.472 to 1 d.p.:

3.4

3.5

3.47

4

9.96 to 1 d.p.:

9.9

10.0

10

9.96

0.0851 to 2 s.f.:

0.08

0.085

0.09

0.0851

205.6 to 2 s.f.:

200

210

205

206

Estimate $38 \cdot 21$:

600

800

1000

80

Estimate $6.9 \cdot 5.2$:

30

35

42

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To round	Yaxlitlash	Округлить
Nearest	Eng yaqin	Ближайший
Decimal place	O‘nli xona	Десятичный знак
Significant figure	Muhim raqam	Значащая цифра
Deciding digit	Hal qiluvchi raqam	Решающая цифра
Estimate	Taxminiy hisob	Оценка
Accuracy	Aniqlik	Точность
Approximately	Taxminan	Приблизительно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A digit of 5 or more means:

round down

round up

leave it

add one to the last digit shown

795 to the nearest ten:

790

800

700

750

3.472 to 2 d.p.:

3.47

3.48

3.5

3.4

In 0.0851 the first significant figure is:

0

8

5

1

$3\,472$ to 1 s.f.:

3 000

3 400

4 000

3 500

An estimate is used to:

get an exact answer

catch a large error

avoid working

round twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3 472 to the nearest ten

ANSWER 3 470

2. 3 472 to the nearest hundred

ANSWER 3 500

3. 2 851 to the nearest thousand

ANSWER 3 000

4. 795 to the nearest ten

ANSWER 800

5. 3.472 to 1 d.p.

ANSWER 3.5

6. 3.472 to 2 d.p.

ANSWER 3.47

7. 2.345 to 2 d.p.

ANSWER 2.35

Medium · the standard question

7 PROBLEMS

1. 9.96 to 1 d.p.

ANSWER 10.0

2. 0.0851 to 2 s.f.

ANSWER 0.085

3. 0.0851 to 1 s.f.

ANSWER 0.09

4. 205.6 to 2 s.f.

ANSWER 210

5. 3 472 to 1 s.f.

ANSWER 3 000

6. Estimate $38 \cdot 21$

ANSWER 800

7. Estimate $297 + 405$

ANSWER 700

Hard · stretch and reason

7 PROBLEMS

1. Estimate $4\,187 \div 21$

ANSWER 200

2. Estimate $6.9 \cdot 5.2$

ANSWER 35

3. 205.6 to 3 s.f.

ANSWER 206

4. Why is 10.0 not the same claim as 10?

ANSWER It claims accuracy to a tenth

5. The smallest whole number that rounds to 700 to the nearest hundred

ANSWER 650

6. The largest whole number that rounds to 700 to the nearest hundred

ANSWER 749

7. A length given as 4.5 m to 1 d.p. lies between

ANSWER 4.45 and 4.55 m

07 Homework

Homework — 5 tasks

Underline the deciding digit before you round anything.

1. Round 6 483 to the nearest ten, hundred and thousand.
2. Round 5.278 to 1 and to 2 decimal places.
3. Round 0.04672 to 1 and to 2 significant figures.
4. Estimate $58 \cdot 42$ and $8\,130 \div 19$.
5. Between which two values does a mass given as 3.2 kg to 1 d.p. lie?

Finding a number from its percentage, and a part of it

The reverse question — given the part and the percentage, find the whole.

LESSON 59–62

4 × 40 MIN

MATEMATIKA 6, §13

S7 10

LESSON CLOCK

25'

35'

50'

40'

10

The three questions	25 min
Finding a percentage	35 min
Finding the whole	50 min
Expressing as a percentage	40 min
Homework	10 min

LEARNING OBJECTIVES

- Find a percentage of a quantity fluently.
- Find the whole when a percentage of it is given.
- Express one quantity as a percentage of another.
- Decide which of the three questions a problem is asking.

TEXTBOOK

National	pp. 167–178
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.2

01

Explanation

The three questions

QUESTION	GIVEN	METHOD	EXAMPLE
find a part	the whole and the percentage	multiply	15% of 240 = 36
find the whole	a part and its percentage	divide	36 is 15% $\implies$ 240
find the percentage	the part and the whole	$\frac{\text{part}}{\text{whole}} \cdot 100$	36 of 240 = 15%

THE SAME THREE NUMBERS, THREE WAYS ROUND

Every percentage problem in the course is one of these. Naming which one is being asked, before calculating anything, is the skill this block is really teaching.

Finding a percentage

TASK	WORKING	ANSWER
30%	$0.3 \cdot 250$	75
8%	$0.08 \cdot 250$	20
45%	$0.45 \cdot 250$	112.5
120%	$1.2 \cdot 250$	300

Every row is a percentage of 250.

TURN THE PERCENTAGE INTO A DECIMAL AND MULTIPLY

One multiplication replaces the two-step “find 1% then multiply”. Both are correct; the decimal method is faster once the conversion is automatic.

Finding the whole

If 36 is 15% of a number, then 1% is $36 \div 15 = 2.4$, so the whole is 240.

$$\text{whole} = \frac{\text{part}}{\text{percentage}} \times 100 \text{ or } \text{part} \div \text{the decimal}$$

GIVEN	1%	WHOLE
36	2.4	240
45	1.5	150
60	3	300
18	0.6	60

THE WHOLE IS LARGER THAN THE PART – UNLESS THE PERCENTAGE IS OVER 100

If 36 is 15% of a number, the number is 240, not 5.4. Multiplying when you should divide is the error, and comparing the two numbers exposes it.

Expressing as a percentage

PART	WHOLE	WORKING	PERCENTAGE
36	240	$\frac{36}{240} \cdot 100$	15%
42	56	$\frac{42}{56} \cdot 100$	75%
7	20	$\frac{7}{20} \cdot 100$	35%
45	36	$\frac{45}{36} \cdot 100$	125%

OVER 100% IS NOT A MISTAKE

The last row compares a larger quantity with a smaller one, so the answer exceeds 100%. It is wrong only when the part really is part of the whole.

02 Worked examples

EXAMPLE 1 Find 45% of 250.

1 $45\% = 0.45$

2 $0.45 \cdot 250$

3 $= 112.5$

ANSWER

112.5

EXAMPLE 2 36 is 15% of a number. Find the number.

- ① 1% is $36 \div 15 = 2.4$.
- ② 100% is 240.
- ③ Check: 15% of 240 is 36 ✓

ANSWER

240

EXAMPLE 3 What percentage of 56 is 42?

- ① $\frac{42}{56}$
The whole underneath.
- ② $= 0.75$
- ③ $\cdot 100 = 75\%$

ANSWER

75%

03 Interactive model

Give the three questions with the same three numbers — 36, 240 and 15% — and ask which is which; the pattern lands in one minute.

Which of the three questions?

“Find 15% of 240” needs:

multiplying

dividing

$\frac{\text{part}}{\text{whole}}$

nothing

“36 is 15% of what?” needs:

multiplying

dividing

adding

squaring

“36 out of 240 as a percentage” needs:

multiplying by 100

dividing then multiplying by 100

dividing by 100

nothing

30% of 250 is:

75

80

7.5

30

8% of 250 is:

2

20

31.25

8

45 is 30% of:

13.5

135

150

75

42 as a percentage of 56:

75%

133%

42%

14%

45 as a percentage of 36:

80%

125%

9%

45%

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Whole	Butun	Целое
Part	Qism	Часть
To find the whole	Butunni topish	Найти целое
One per cent	Bir foiz	Один процент
Reverse question	Teskari masala	Обратная задача
To express as	Ifodalamoq	Выразить в
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To find a percentage of a quantity you:

To find the whole from a part you:

36 is 15% of:

120% of 250 is:

270

300

320

250

7 as a percentage of 20:

7%

35%

14%

70%

A percentage over 100 means:

an error

the part exceeds the whole compared with

a fraction

a decimal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *30% of 250*

ANSWER *75*

2. *8% of 250*

ANSWER *20*

3. *45% of 250*

ANSWER *112.5*

4. *120% of 250*

ANSWER *300*

5. *36 is 15% of*

ANSWER *240*

6. *45 is 30% of*

ANSWER *150*

7. *60 is 20% of*

ANSWER *300*

Medium · the standard question

7 PROBLEMS

1. 18 is 30% of

ANSWER 60

2. 36 as a percentage of 240

ANSWER 15%

3. 42 as a percentage of 56

ANSWER 75%

4. 7 as a percentage of 20

ANSWER 35%

5. 45 as a percentage of 36

ANSWER 125%

6. 12% of 4 500

ANSWER 540

7. 84 is 70% of

ANSWER 120

Hard · stretch and reason

7 PROBLEMS

1. 2.5% of 4 800

ANSWER 120

2. 150 is 12.5% of

ANSWER 1 200

3. A price of 54 000 is 90% of the original: the original

ANSWER 60 000

4. 64 as a percentage of 80

ANSWER 80%

5. If 30% of a class of 40 wear glasses, how many do not?

ANSWER 28

6. A tax of 12% adds 36 000: the price before tax

ANSWER 300 000

7. Why must the whole go underneath?

ANSWER A percentage compares the part with the whole it belongs to

07 Homework

Homework — 5 tasks

Name which of the three questions you are answering before you calculate.

1. Find 35% and 6% of 420.
2. 54 is 18% of a number. Find the number.
3. Write 27 as a percentage of 45.
4. Write 63 as a percentage of 45, and explain the answer.
5. A sum of 84 000 is 70% of a price. Find the price.

Finding a percentage increase or decrease

One multiplier does the whole job — and the reverse question needs a division.

LESSON 63–66

4 × 40 MIN

MATEMATIKA 6, §14

S7 10

LESSON CLOCK

30'

40'

50'

30'

10

The multiplier	30 min
Finding the change	40 min
The reverse question	50 min
Profit, loss and discount	30 min
Homework	10 min

LEARNING OBJECTIVES

- Increase and decrease a quantity by a percentage.
- Find the percentage change from the old and new values.
- Find the original value after a change.
- Explain why a rise and a fall of the same percentage do not cancel.

TEXTBOOK

National	pp. 179–190
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.3

01

Explanation

The multiplier

new = old × (1 ± $\frac{P}{100}$)

CHANGE	MULTIPLIER	80 000	250
up 20%	1.2	96 000	300
down 20%	0.8	64 000	200
up 7%	1.07	85 600	267.5
down 35%	0.65	52 000	162.5

ONE STEP, NOT TWO

Finding 20% and then adding it works, but · 1.2 is one keystroke and cannot lose the original amount along the way.

Finding the change

percentage change = $\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$

OLD	NEW	CHANGE	PERCENTAGE
40	50	+10	+25%
50	40	−10	−20%
80 000	92 000	+12 000	+15%
200	150	−50	−25%

ALWAYS DIVIDE BY THE OLD VALUE

The first two rows use the same pair of numbers and give different percentages, because the base changes. A rise of 25% is undone by a fall of 20%, not of 25%.

The reverse question

“After a rise of 20% the price is 96 000” — the 96 000 is the **new** value, so divide by the multiplier.

GIVEN	WRONG	RIGHT	ANSWER
after +20% it is 96 000	96 000 − 19 200	96 000 ÷ 1.2	80 000
after −20% it is 64 000	64 000 + 12 800	64 000 ÷ 0.8	80 000
after −25% it is 150	150 + 37.5	150 ÷ 0.75	200

CHECK BY GOING FORWARDS

80 000 · 1.2 = 96 000 ✓. The forward check takes five seconds and settles whether the division was the right move.

Profit, loss and discount

SITUATION	WORKING	ANSWER
bought at 40 000, sold at 50 000	$\frac{10\,000}{40\,000} \cdot 100$	25% profit
bought at 50 000, sold at 40 000	$\frac{10\,000}{50\,000} \cdot 100$	20% loss
a coat at 300 000 with 30% off	$\cdot 0.7$	210 000
a bill of 120 000 plus 12% tax	$\cdot 1.12$	134 400

PROFIT AND LOSS ARE PERCENTAGES OF THE COST

A trader who buys at 40 000 and sells at 50 000 makes 25%, not 20% — the cost price is the base. Using the selling price instead is the classic error.

02 Worked examples

EXAMPLE 1 Increase 80 000 by 15%.

1 Multiplier: 1.15.

2 $80\,000 \cdot 1.15$

3 $= 92\,000$

ANSWER

92 000

EXAMPLE 2 A price falls from 200 to 150. Find the percentage decrease.

① Change: -50 .

② $\frac{50}{200} \cdot 100$

The old value underneath.

③ $= 25\%$ decrease.

ANSWER

25%

EXAMPLE 3 After a fall of 20% a price is 64 000. What was it before?

① The multiplier was 0.8.

② $64\,000 \div 0.8$

Divide, do not add.

③ $= 80\,000$

Check: $80\,000 \cdot 0.8 = 64\,000 \checkmark$

ANSWER

80 000

03 Interactive model

Ask whether a shop that raises a price 20% and then takes 20% off has returned to the start; the near-unanimous “yes” is worth correcting in public.

Multiplier, change, or reverse?

The multiplier for a rise of 15%:

The multiplier for a fall of 35%:

80 000 up 15%:

From 40 to 50:

+20%

+25%

+10%

+50%

From 50 to 40:

-20%

-25%

-10%

-80%

After +20% a price is 96 000; before:

76 800

80 000

115 200

86 400

Bought at 40 000, sold at 50 000:

20% profit

25% profit

10% profit

25% loss

Up 20% then down 20% gives:

the same price

96% of it

104% of it

90% of it

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Multiplier	Ko'paytuvchi	Множитель
Percentage change	Foizli o'zgarish	Процентное изменение
Original value	Boshlang'ich qiymat	Первоначальное значение
Discount	Chegirma	Скидка
Profit	Foyda	Прибыль
Loss	Zarar	Убыток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To increase by 7% multiply by:

To decrease by 35% multiply by:

Percentage change divides by:

After -25% a price is 150; before:

187.5

200

175

112.5

A rise of 25% is undone by a fall of:

25%

20%

30%

75%

Profit is a percentage of:

the selling price

the cost price

the profit

the tax

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 80 000 up 20%

ANSWER 96 000

2. 80 000 down 20%

ANSWER 64 000

3. 250 up 7%

ANSWER 267.5

4. 250 down 35%

ANSWER 162.5

5. The multiplier for +15%

ANSWER 1.15

6. The multiplier for -35%

ANSWER 0.65

7. 300 000 with 30% off

ANSWER 210 000

Medium · the standard question

7 PROBLEMS

1. From 40 to 50

ANSWER +25%

2. From 50 to 40

ANSWER -20%

3. From 80 000 to 92 000

ANSWER +15%

4. From 200 to 150

ANSWER -25%

5. After +20% it is 96 000: before

ANSWER 80 000

6. After -20% it is 64 000: before

ANSWER 80 000

7. A bill of 120 000 plus 12% tax

ANSWER 134 400

Hard · stretch and reason

7 PROBLEMS

1. Bought at 40 000, sold at 50 000: the profit percentage

ANSWER 25%

2. Bought at 50 000, sold at 40 000: the loss percentage

ANSWER 20%

3. Up 20% then down 20% from 500

ANSWER 480

4. A price after two rises of 10% from 200

ANSWER 242

5. After -40% a price is 72 000: before

ANSWER 120 000

6. A coat costs 210 000 after 30% off: the original

ANSWER 300 000

7. Why is a 25% rise undone by a 20% fall?

ANSWER The fall is taken from a larger amount

07 Homework

Homework — 5 tasks

Every reverse question is a division; check it by multiplying forwards.

1. Increase 45 000 by 16% and decrease it by 16%.
2. A price rises from 250 to 310. Find the percentage increase.
3. After a fall of 25% a price is 180 000. Find the original price.
4. A trader buys at 60 000 and sells at 75 000. Find the profit percentage.
5. Show that a rise of 50% followed by a fall of 50% leaves 75% of the original.

Word problems on percentages

Shops, salaries, populations and exam marks — all the same three questions in disguise.

LESSON 67–70

4 × 40 MIN

MATEMATIKA 6, §15

S7 10

LESSON CLOCK

30'

40'

50'

30'

10

Naming the question	30 min
Shops and money	40 min
Successive changes	50 min
Populations and marks	30 min
Homework	10 min

LEARNING OBJECTIVES

- Choose which percentage question a problem is asking.
- Work with successive percentage changes.
- Solve problems about discounts, taxes and interest.
- Interpret a percentage answer in the context it came from.

TEXTBOOK

National	pp. 191–202
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.4

01

Explanation

Naming the question

WORDING	QUESTION	OPERATION
“find 15% of...”	a part	multiply
“...is 15% of what?”	the whole	divide
“what percentage of... is...?”	the percentage	$\frac{\text{part}}{\text{whole}} \cdot 100$
“increase by 15%”	a new value	$\cdot 1.15$
“after a rise of 15% it is...”	the original	$\div 1.15$

FIVE PHRASES, FIVE OPERATIONS

Nearly every percentage problem in the paper opens with one of those five wordings. Matching the phrase to the operation is more reliable than reasoning from the numbers.

Shops and money

PROBLEM	WORKING	ANSWER
a jacket at 400 000 with 25% off	$\cdot 0.75$	300 000
a bill of 80 000 plus 15% service	$\cdot 1.15$	92 000
a salary of 2 400 000 rising 8%	$\cdot 1.08$	2 592 000
1 000 000 at 14% simple interest for a year	$\cdot 0.14$	140 000 interest
a sale price of 300 000 after 25% off	$\div 0.75$	400 000

THE LAST ROW IS THE FIRST ROW BACKWARDS

Shops advertise the discount and the sale price; finding the original from them is the question most likely to appear in an examination.

Successive changes

Two changes multiply — they never add.

CHANGES	COMBINED MULTIPLIER	NET CHANGE
up 20%, then up 10%	$1.2 \cdot 1.1 = 1.32$	+32%
up 20%, then down 20%	$1.2 \cdot 0.8 = 0.96$	−4%
down 10% twice	$0.9 \cdot 0.9 = 0.81$	−19%
up 50%, then down 50%	$1.5 \cdot 0.5 = 0.75$	−25%

+20% THEN +10% IS NOT +30%

It is +32%, because the second rise is taken from the already-increased amount. Adding percentage changes is the commonest error in this whole topic.

Populations and marks

PROBLEM	WORKING	ANSWER
a village of 2 500 grows 4%	$\cdot 1.04$	2 600
a mark of 34 out of 40	$\frac{34}{40} \cdot 100$	85%
18 of 24 pupils passed	$\frac{18}{24} \cdot 100$	75%
attendance rose from 80% to 92%	$92 - 80$	12 percentage points

PERCENTAGE POINTS ARE NOT PERCENTAGES

A rise from 80% to 92% is 12 percentage points — but a 15% increase in the attendance figure. Newspapers confuse the two constantly; a mathematician should not.

02

Worked examples

EXAMPLE 1 A jacket costing 400 000 sum is reduced by 25%. Find the sale price.

1

A fall of 25% means multiplying by 0.75.

2

$400\,000 \cdot 0.75$

3

$= 300\,000$ sum.

ANSWER

300 000 sum

EXAMPLE 2 A price rises 20% and then falls 20%. What is the net change?

- ① Multipliers: 1.2 and 0.8.
- ② $1.2 \cdot 0.8 = 0.96$
- ③ That is 96% of the original — a fall of 4%.

ANSWER

-4%

EXAMPLE 3 A pupil scores 34 out of 40. Give the mark as a percentage.

- ① $\frac{34}{40}$
- ② $= 0.85$
- ③ 85%

ANSWER

85%

03 Interactive model

Collect real advertisements with percentages on them; the class checks the arithmetic and often finds the offer is worse than it looks.

Which question, and what answer?

400 000 with 25% off:

100 000

300 000

375 000

500 000

A bill of 80 000 plus 15%:

92 000

95 000

12 000

68 000

A sale price of 300 000 after 25% off; the original:

375 000

400 000

325 000

225 000

Up 20% then up 10% is:

+30%

+32%

+2%

+22%

Up 20% then down 20% is:

0%

-4%

+4%

-40%

Down 10% twice is:

-20%

-19%

-21%

-100%

34 out of 40 as a percentage:

80%

85%

75%

88%

From 80% to 92% is:

12 percentage points

12% increase

both the same

92%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Discount	Chegirma	Скидка
Tax	Soliq	Налог
Interest	Foiz (bank)	Проценты
Salary	Maosh	Зарплата
Population	Aholi	Население
Successive change	Ketma-ket o'zgarish	Последовательное изменение
Net change	Yakuniy o'zgarish	Итоговое изменение
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Increase by 15%” means multiply by:

0.15

1.15

15

0.85

“After a rise of 15% it is...” means:

multiply by 1.15

divide by 1.15

subtract 15%

add 15%

Two successive changes:

add

multiply

subtract

cancel

A salary of 2 400 000 rising 8% becomes:

2 592 000

2 592

2 400 008

2 208 000

1 000 000 at 14% for a year earns:

14 000

140 000

1 400 000

114 000

A rise from 80% to 92% is an increase of:

12%

15%

92%

8%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 400 000 with 25% off

ANSWER 300 000

2. 80 000 plus 15%

ANSWER 92 000

3. 2 400 000 up 8%

ANSWER 2 592 000

4. 1 000 000 at 14% for a year: the interest

ANSWER 140 000

5. 2 500 up 4%

ANSWER 2 600

6. 34 out of 40

ANSWER 85%

7. 18 out of 24

ANSWER 75%

Medium · the standard question

7 PROBLEMS

1. A sale price of 300 000 after 25% off: the original

ANSWER 400 000

2. Up 20% then up 10%

ANSWER +32%

3. Up 20% then down 20%

ANSWER -4%

4. Down 10% twice

ANSWER -19%

5. Up 50% then down 50%

ANSWER -25%

6. From 80% to 92%: the increase

ANSWER 15%

7. A price of 92 000 after 15% was added: before

ANSWER 80 000

Hard · stretch and reason

7 PROBLEMS

1. A shop adds 20% then advertises 20% off: the net change

ANSWER -4%

2. A population of 2 500 after two years at 4% a year

ANSWER 2 704

3. A price falls 30% to 84 000: the original

ANSWER 120 000

4. Which is better: 30% off, or 20% off then 10% more off?

ANSWER 30% off — the other leaves 72%

5. A mark rose from 60 to 75 out of 100: the percentage increase

ANSWER 25%

6. A salary rises 10% then falls 10%: the net change

ANSWER -1%

7. Why do successive percentages multiply?

ANSWER Each acts on the amount left by the one before

07 Homework

Homework — 5 tasks

Write the multiplier for every change before you touch the numbers.

1. A coat costs 520 000 sum and is reduced by 35%. Find the sale price.
2. A bill of 140 000 has 12% added. Find the total.
3. After a discount of 40% a price is 96 000. Find the original price.
4. A price rises 25% and then falls 20%. Find the net change.
5. A pupil scores 57 out of 60. Give the mark as a percentage.

Think — Project

2: clock rectangles

A Cambridge project: the numbers on a clock face, boxed four at a time, and the pattern that appears.

LESSON 71

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 PROJECT 2

LESSON CLOCK

8'

12'

14'

6'

The arrangement	8 min
Collecting the data	12 min
Stating the pattern	14 min
Explaining why	6 min

LEARNING OBJECTIVES

- Collect data systematically from a simple arrangement.
- Notice a pattern and state it in words.
- Express the pattern with letters.
- Test the statement on cases not yet tried.

TEXTBOOK

National	pp. 203–205
Cambridge	Stage 7, project pages
Workbook	Project sheet 2

01

Explanation

The arrangement

Write the numbers 1 to 12 in three rows of four, as on a calendar page:

1	2	3	4
5	6	7	8
9	10	11	12

Draw a rectangle round any four numbers that form a block — two rows and two columns — and multiply each pair of opposite corners.

THE QUESTION

What is the difference between the two products? Try several rectangles before deciding.

Collecting the data

RECTANGLE	ONE DIAGONAL	THE OTHER	DIFFERENCE
1, 2, 5, 6	$1 \cdot 6 = 6$	$2 \cdot 5 = 10$	4
2, 3, 6, 7	$2 \cdot 7 = 14$	$3 \cdot 6 = 18$	4
6, 7, 10, 11	$6 \cdot 11 = 66$	$7 \cdot 10 = 70$	4
3, 4, 7, 8	$3 \cdot 8 = 24$	$4 \cdot 7 = 28$	4

BE SYSTEMATIC, NOT RANDOM

Working across the top row and then down catches every case and makes it obvious when the pattern holds. Random examples leave gaps that hide exceptions.

Stating the pattern

The difference is always 4. Now say why, with letters: let the top-left number be n .

POSITION	NUMBER
top left	n
top right	$n + 1$
bottom left	$n + 4$
bottom right	$n + 5$

$$(n + 1)(n + 4) - n(n + 5) = n^2 + 5n + 4 - n^2 - 5n = 4$$

THE LETTERS PROVE WHAT THE NUMBERS SUGGESTED

Four examples show a pattern; the algebra shows it must hold for every rectangle in the grid, including ones nobody has tried.

Explaining why

GRID WIDTH	BOTTOM-LEFT IS	THE DIFFERENCE
4	$n + 4$	4
5	$n + 5$	5
7	$n + 7$	7
w	$n + w$	w

THE DIFFERENCE IS THE WIDTH OF THE GRID

A calendar month, seven columns wide, gives a difference of 7 for every 2×2 block — which is worth testing on a real calendar before the lesson ends.

02

Worked examples

EXAMPLE 1 Take the rectangle 6, 7, 10, 11. Find the two products and their difference.

1

$6 \cdot 11 = 66$

2

$7 \cdot 10 = 70$

3

$70 - 66 = 4$

ANSWER

4

EXAMPLE 2 Prove that the difference is always 4 in a grid four columns wide.

- 1 Let the top-left number be n .
- 2 The four numbers are $n, n + 1, n + 4, n + 5$.
- 3 $(n + 1)(n + 4) = n^2 + 5n + 4$
- 4 $n(n + 5) = n^2 + 5n$, so the difference is 4.

ANSWER

Always 4

EXAMPLE 3 What would the difference be on a calendar, seven columns wide?

- 1 The bottom-left number is $n + 7$.
- 2 $(n + 1)(n + 7) = n^2 + 8n + 7$
- 3 $n(n + 8) = n^2 + 8n$
- 4 The difference is 7.

ANSWER

7

03 Interactive model

Hand out real calendar pages and let the class discover the seven before any algebra; the general result then answers a question they have already asked.

The clock-rectangle pattern

For 1, 2, 5, 6 the difference is:

For 6, 7, 10, 11 the difference is:

If the top left is n , the bottom right is:

$(n + 1)(n + 4)$ equals:

$$n^2 + 4$$

$$n^2 + 5n + 4$$

$$n^2 + 4n + 1$$

$$n^2 + 5$$

$n(n + 5)$ equals:

$$n^2 + 5$$

$$n^2 + 5n$$

$$n + 5n$$

$$5n^2$$

So the difference is:

$$n$$

$$4$$

$$5$$

$$5n$$

On a seven-column calendar the difference is:

$$4$$

$$7$$

$$12$$

$$49$$

Four examples showing the pattern are:

a proof

evidence for a conjecture

irrelevant

a counter-example

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Clock face	Soat siferblati	Циферблат
Rectangle	To'rtburchak	Прямоугольник
Diagonal	Diagonal	Диагональ
Pattern	Qonuniyat	Закономерность
Conjecture	Faraz	Гипотеза
To generalise	Umumlashtirish	Обобщить
Systematic	Tartibli	Систематический
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In the 3×4 grid the difference is always:

With top left n , the bottom left is:

The proof needs:

On a grid w wide the difference is:

4

w

$2w$

w^2

A pattern seen in four cases is:

proved

a conjecture

false

a theorem

Systematic data collection means:

random examples

working in order

one example

guessing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The difference for 1, 2, 5, 6

ANSWER 4

2. The difference for 2, 3, 6, 7

ANSWER 4

3. The difference for 3, 4, 7, 8

ANSWER 4

4. The difference for 6, 7, 10, 11

ANSWER 4

5. With top left n , the top right

ANSWER $n + 1$

6. The bottom left

ANSWER $n + 4$

7. The bottom right

ANSWER $n + 5$

Medium · the standard question

7 PROBLEMS

1. $(n + 1)(n + 4)$ expanded

ANSWER $n^2 + 5n + 4$

2. $n(n + 5)$ expanded

ANSWER $n^2 + 5n$

3. Their difference

ANSWER 4

4. On a five-column grid

ANSWER 5

5. On a calendar (seven columns)

ANSWER 7

6. On a grid w columns wide

ANSWER w

7. Is the pattern proved by four examples?

ANSWER No — they are evidence only

Hard · stretch and reason

7 PROBLEMS

1. A 3×3 block on a seven-column calendar: the corner difference

ANSWER 28

2. Why does the n^2 term always cancel?

ANSWER Both products have the same leading term

3. Predict the difference on a ten-column grid

ANSWER 10

4. Test the prediction with $n = 3$ on a ten-column grid

ANSWER $4 \cdot 13 - 3 \cdot 14 = 10$ ✓

5. What would happen with a 2×3 block on a four-column grid?

ANSWER The difference becomes 8

6. Write the four numbers of a block with bottom right n

ANSWER $n - 5, n - 4, n - 1, n$

7. Why is algebra better than more examples here?

ANSWER It settles every case at once, including untried ones

07 Homework

Homework — the project

One page: your data table, the pattern in words, the algebra, and one prediction tested.

1. Draw the 3×4 grid and test six different rectangles.
2. Record your results in a table and state the pattern in one sentence.
3. Prove the pattern using n for the top-left number.
4. Repeat the investigation on a real calendar month and record what happens.
5. Predict the difference on a grid 9 columns wide, then test your prediction.

Revision — percentages

The three questions, the multiplier and the reverse case, gathered before the control work.

LESSON 72–73

2 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 10 CONSOLIDATION

LESSON CLOCK

20'

25'

25'

8'

2

The methods on one page

20 min

Worked mixture

25 min

The errors that remain

25 min

A last check

8 min

Homework

2 min

LEARNING OBJECTIVES

- Choose the right method for each kind of percentage question.
- Work with increases, decreases and successive changes.
- Solve the reverse question reliably.
- Identify the errors that remain.

TEXTBOOK

National

pp. 206–210

Cambridge

Stage 7, pp. 96–104

Workbook

Revision 10

01

Explanation

The methods on one page

QUESTION	METHOD	EXAMPLE
a percentage of an amount	multiply by the decimal	$0.35 \cdot 240 = 84$
the whole from a part	divide by the decimal	$84 \div 0.35 = 240$
one amount as a percentage of another	$\frac{\text{part}}{\text{whole}} \cdot 100$	$\frac{84}{240} \cdot 100 = 35\%$
increase or decrease	$\cdot (1 \pm \frac{p}{100})$	$240 \cdot 1.35 = 324$
the original after a change	divide by the multiplier	$324 \div 1.35 = 240$
successive changes	multiply the multipliers	$1.2 \cdot 0.8 = 0.96$

SIX ROWS, AND THE WHOLE TOPIC

Every percentage question in the paper is one of these. Read the wording, pick the row, do the one operation it names.

Worked mixture

PROBLEM	ROW	ANSWER
18% of 350	first	63
63 is 18% of what?	second	350
63 as a percentage of 350	third	18%
increase 350 by 18%	fourth	413
after +18% it is 413: before	fifth	350
up 18%, then down 18%	sixth	-3.24%

ONE SET OF NUMBERS, ALL SIX QUESTIONS

Seeing 63, 350 and 18% arranged six ways is the fastest way to stop confusing the questions with each other.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
dividing by the new value	$\frac{10}{50}$ for $40 \rightarrow 50$	$\frac{10}{40} = 25\%$
subtracting instead of dividing	$96\,000 - 19\,200$	$96\,000 \div 1.2$
adding successive percentages	$+20\% + 10\% = +30\%$	$+32\%$
a percentage over 100 rejected	“that must be wrong”	it is possible
profit taken on the selling price	$\frac{10\,000}{50\,000}$	$\frac{10\,000}{40\,000}$

A last check

ANSWER	SENSE CHECK
a part	smaller than the whole, unless the percentage is over <i>100</i>
the whole from a part	larger than the part
after an increase	larger than before
after a decrease	smaller than before
a reverse question	multiply forwards and see

THE SENSE CHECK COSTS NOTHING

Most lost marks in this topic are answers that are obviously the wrong size. Ten seconds of checking recovers them.

02 Worked examples

EXAMPLE 1 Find *18%* of *350*, and then check that *63* is *18%* of *350*.

① $0.18 \cdot 350 = 63$

② $\frac{63}{350} = 0.18$

③ $18\% \checkmark$

The first and third questions, one after the other.

ANSWER

63 and *18%*

EXAMPLE 2 After a rise of 18% a price is 413. Find the original.

① Multiplier 1.18.

② $413 \div 1.18$

③ $= 350$

Check: $350 \cdot 1.18 = 413 \checkmark$

ANSWER

350

EXAMPLE 3 A price goes up 18% and then down 18%. What is the net change?

① $1.18 \cdot 0.82$

② $= 0.9676$

③ A fall of 3.24%.

Never zero.

ANSWER

−3.24%

03 Interactive model

Write 63, 350 and 18% on the board and ask the class to invent all six questions; they revise the topic by writing it rather than reading it.

Which row of the table?

18% of 350:

63

350

413

18

63 is 18% of:

11.34

350

413

81

63 as a percentage of 350:

18%

56%

5.6%

180%

350 increased by 18%:

368

413

287

63

After +18% it is 413; before:

339

350

487

395

Up 18% then down 18%:

0%

-3.24%

+3.24%

-36%

From 40 to 50 the base is:

40

50

10

90

A profit percentage uses:

the selling price

the cost price

the profit

either

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Multiplier	Ko'paytuvchi	Множитель
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Original value	Boshlang'ich qiymat	Первоначальное значение
Percentage change	Foizli o'zgarish	Процентное изменение
Successive	Ketma-ket	Последовательный
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A percentage of an amount:

multiply

divide

add

subtract

The whole from a part:

multiply

divide

add

subtract

The original after an increase:

multiply by the multiplier

divide by the multiplier

subtract the percentage

add it back

Successive changes:

add

multiply

cancel

subtract

Percentage change divides by:

the new value

the old value

100

the change

An answer larger than the whole is:

always wrong

possible

impossible

a decimal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *18% of 350*

ANSWER 63

2. *63 is 18% of*

ANSWER 350

3. *63 as a percentage of 350*

ANSWER 18%

4. *350 up 18%*

ANSWER 413

5. *350 down 18%*

ANSWER 287

6. *The multiplier for +18%*

ANSWER 1.18

7. *The multiplier for −18%*

ANSWER 0.82

Medium · the standard question

7 PROBLEMS

1. After +18% it is 413: before

ANSWER 350

2. After -18% it is 287: before

ANSWER 350

3. Up 18% then down 18%

ANSWER -3.24%

4. From 40 to 50

ANSWER +25%

5. From 50 to 40

ANSWER -20%

6. Bought 40 000, sold 50 000

ANSWER 25% profit

7. 12.5% of 480

ANSWER 60

Hard · stretch and reason

7 PROBLEMS

1. A price after 30% off is 147 000: the original

ANSWER 210 000

2. Up 25% then down 20%

ANSWER No change

3. Two rises of 15% from 200

ANSWER 264.5

4. 72 as a percentage of 60

ANSWER 120%

5. A salary of 3 000 000 after 12% rise and 5% tax

ANSWER 3 192 000

6. Which loses more: –10% twice, or –20% once?

ANSWER –20% once, at 0.8 against 0.81

7. Why check the size of a percentage answer?

ANSWER Most lost marks are answers of obviously the wrong size

07 Homework

Homework — 5 tasks

Name the row of the method table before every calculation.

- Find 22% of 450 and write 99 as a percentage of 450.
- Increase 450 by 22% and decrease it by 22%.
- After a rise of 22% a price is 549. Find the original.
- A price rises 30% and then falls 30%. Find the net change.
- A trader buys at 80 000 and sells at 92 000. Find the profit percentage.

Control work 4 — percentages, and work on the mistakes

The three questions, the multiplier and the reverse case, under time.

LESSON 74–75

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 4

S7 10 REVIEW

LESSON CLOCK

3

35'

12'

25'

5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Convert between percentages, fractions and decimals.
- Find a part, a whole and a percentage.
- Handle an increase, a decrease and a reverse question.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 154–210
Cambridge	Stage 7, pp. 96–104
Workbook	Control paper 4

01

Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	Write 35% as a fraction in lowest terms and as a decimal	3	L55
2	Find 18% of 350	3	L59–62
3	63 is 18% of a number: find it	4	L59–62
4	Write 42 as a percentage of 56	3	L59–62
5	Increase 80 000 by 15%	3	L63–66
6	After a fall of 20% a price is 64 000: find the original	4	L63–66
7	A trader buys at 40 000 and sells at 50 000: find the profit percentage	5	L67–70

THE ANSWERS

$\frac{7}{20}$ and 0.35; 63; 350; 75%; 92 000; 80 000; 25% profit.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
fraction not simplified	$\frac{35}{100}$	$\frac{7}{20}$
decimal point misplaced	3.5	0.35
multiplied instead of divided	$63 \cdot 0.18$	$63 \div 0.18$
part and whole swapped	$\frac{56}{42}$	$\frac{42}{56}$
increase found but not added	12 000	92 000
reverse question added back	$64\,000 + 12\,800$	$64\,000 \div 0.8$
profit on the selling price	$\frac{10\,000}{50\,000} = 20\%$	$\frac{10\,000}{40\,000} = 25\%$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

QUESTION	OPERATION	SENSE CHECK
a part	multiply	smaller than the whole
the whole	divide	larger than the part
the percentage	$\frac{\text{part}}{\text{whole}} \cdot 100$	under 100 if it really is a part
increase	$\cdot (1 + \frac{p}{100})$	larger
decrease	$\cdot (1 - \frac{p}{100})$	smaller
reverse	divide by the multiplier	multiply forwards to check

LOOKING FORWARD

Percentages return in the pie chart in Quarter IV, where 360° is shared out, and in every later problem about money. The next two lessons are the decimals that all this arithmetic needs.

02 Worked examples

EXAMPLE 1 Model answer, Q3: 63 is 18% of a number.

① The whole is wanted, so divide.

② $63 \div 0.18$

③ $= 350$

④ Check: $0.18 \cdot 350 = 63 \checkmark$

ANSWER

350

EXAMPLE 2 Model answer, Q6: after -20% a price is 64 000.

① The multiplier was 0.8.

② $64\,000 \div 0.8$

Divide, never add back.

③ $= 80\,000$

Check: $80\,000 \cdot 0.8 = 64\,000 \checkmark$

ANSWER

80 000

EXAMPLE 3 Model answer, Q7: bought at 40 000, sold at 50 000.

① Profit: 10 000.

② The base is the cost price.

③ $\frac{10\,000}{40\,000} \cdot 100$

④ = 25%

ANSWER

25% profit

03 Interactive model

Return Q6 and Q7 first; both are decided by which number is the base, and naming the base is what the class needs to practise.

The chapter in eight questions

35% as a fraction in lowest terms:

$$\frac{35}{100}$$

$$\frac{7}{20}$$

$$\frac{1}{3}$$

$$\frac{3}{5}$$

35% as a decimal:

3.5

0.35

0.035

35

18% of 350:

63

68

54

35

63 is 18% of:

11.34

350

81

413

42 as a percentage of 56:

75%

133%

42%

24%

80 000 increased by 15%:

92 000

12 000

80 015

68 000

After -20% it is 64 000; before:

76 800

80 000

51 200

84 000

Bought 40 000, sold 50 000:

20%

25%

10%

125%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Percentage	Foiz	Процент
Multiplier	Ko'paytuvchi	Множитель
Whole	Butun	Целое
Increase	Ortish	Увеличение
Original value	Boshlang'ich qiymat	Первоначальное значение
Profit	Foyda	Прибыль
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs the fraction:

as written

in lowest terms

as a decimal only

as a percentage

Q3 is:

multiplication

division

addition

a percentage

Q4 puts underneath:

42

56

100

14

Q5 ends with:

the increase

the new amount

the percentage

the multiplier

Q6 is:

multiplication

division

subtraction

addition

Q7 divides by:

the selling price

the cost price

the profit

100

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 35% as a fraction

ANSWER $\frac{7}{20}$

2. 35% as a decimal

ANSWER 0.35

3. 18% of 350

ANSWER 63

4. 63 is 18% of

ANSWER 350

5. 42 as a percentage of 56

ANSWER 75%

6. 80 000 up 15%

ANSWER 92 000

7. After -20% it is 64 000: before

ANSWER 80 000

Medium · the standard question

7 PROBLEMS

1. Bought 40 000, sold 50 000

ANSWER 25% profit

2. 12% of 2 500

ANSWER 300

3. 96 is 40% of

ANSWER 240

4. 27 as a percentage of 45

ANSWER 60%

5. 600 down 35%

ANSWER 390

6. After +25% it is 500: before

ANSWER 400

7. Bought 60 000, sold 54 000

ANSWER 10% loss

Hard · stretch and reason

7 PROBLEMS

1. Up 20% then down 25%

ANSWER -10%

2. A price after 40% off is 66 000: the original

ANSWER 110 000

3. 84 as a percentage of 70

ANSWER 120%

4. A salary rises 8% to 2 700 000: the old salary

ANSWER 2 500 000

5. 2.5% of 16 000

ANSWER 400

6. Which is worth more: 15% of 80 or 80% of 15?

ANSWER Equal — both 12

7. Why is the base always named first?

ANSWER Every percentage is a percentage of something

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write 64% as a fraction in lowest terms and as a decimal.
3. Find 24% of 750, and say what 180 is as a percentage of 750.
4. After a rise of 30% a price is 91 000. Find the original.
5. A trader buys at 120 000 and sells at 138 000. Find the profit percentage.

Ordering, adding and subtracting decimals

A Cambridge insert: place value decides the order, and the points line up for the arithmetic.

LESSON 76–77

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 4.1–4.2

LESSON CLOCK

20'

25'

25'

8'

2

Ordering	20 min
Adding	25 min
Subtracting	25 min
Estimating	8 min
Homework	2 min

LEARNING OBJECTIVES

- Compare and order decimals of different lengths.
- Add and subtract decimals by lining up the points.
- Use zeros to fill empty places.
- Estimate a decimal calculation before doing it.

TEXTBOOK

National	pp. 211–216
Cambridge	Stage 7, pp. 42–48
Workbook	Exercise 4.1

01

Explanation

Ordering

Compare column by column from the left, not by counting digits. Fill the short numbers with trailing zeros so every number has the same length.

NUMBERS	FILLED OUT	ORDER
0.7, 0.65	0.70, 0.65	0.65 < 0.7
3.4, 3.25, 3.5	3.40, 3.25, 3.50	3.25 < 3.4 < 3.5
0.09, 0.1	0.09, 0.10	0.09 < 0.1

MORE DIGITS DOES NOT MEAN BIGGER

0.65 has more digits than 0.7 and is smaller. The comparison is made column by column: tenths first, then hundredths.

Adding

Write the numbers with their points in a vertical line and add column by column, exactly as with whole numbers.

SUM	LINED UP	ANSWER
$4.7 + 2.85$	$4.70 + 2.85$	7.55
$12.6 + 0.47$	$12.60 + 0.47$	13.07
$3 + 0.45$	$3.00 + 0.45$	3.45
$0.8 + 0.9$	$0.8 + 0.9$	1.7

A WHOLE NUMBER HAS A POINT TOO

3 is 3.00 when it is being added to hundredths. Writing the zeros makes the columns line up and the arithmetic ordinary.

Subtracting

DIFFERENCE	LINED UP	ANSWER
$9.1 - 3.62$	$9.10 - 3.62$	5.48
$5 - 1.35$	$5.00 - 1.35$	3.65
$7.4 - 0.85$	$7.40 - 0.85$	6.55
$0.5 - 0.28$	$0.50 - 0.28$	0.22

FILL THE GAP BEFORE BORROWING

$9.1 - 3.62$ needs the 9.1 written as 9.10; without the zero there is nothing in the hundredths column to borrow from, and the answer comes out wrong by a tenth.

Estimating

CALCULATION	ESTIMATE	EXACT
$4.7 + 2.85$	$5 + 3 = 8$	7.55
$9.1 - 3.62$	$9 - 4 = 5$	5.48
$12.6 + 0.47$	$13 + 0 = 13$	13.07

THE ESTIMATE PLACES THE POINT

An answer of 75.5 or 0.755 would be rejected at once by an estimate of 8. A misplaced decimal point is the error estimating is best at catching.

02 Worked examples

EXAMPLE 1 Order 3.4, 3.25 and 3.5.

- 1

Fill out: 3.40, 3.25, 3.50.
- 2

Compare the tenths: $2 < 4 < 5$.
- 3

$3.25 < 3.4 < 3.5$

ANSWER

$3.25 < 3.4 < 3.5$

EXAMPLE 2 Compute $9.1 - 3.62$.

- 1 Write 9.1 as 9.10 .
- 2 Line up the points and subtract.
- 3 $= 5.48$
Estimate: $9 - 4 = 5$ ✓

ANSWER 5.48 **EXAMPLE 3** Compute $3 + 0.45$ and $5 - 1.35$.

- 1 $3.00 + 0.45 = 3.45$
- 2 $5.00 - 1.35 = 3.65$
Whole numbers get zeros too.

ANSWER 3.45 and 3.65 **03** **Interactive model**

Ask which is bigger, 0.7 or 0.65 , before teaching anything; the split in the class shows exactly who is counting digits rather than reading place value.

Place value decides

Which is larger, 0.7 or 0.65 ?

0.7

0.65

equal

cannot say

Which is larger, 0.09 or 0.1 ?

0.09

0.1

equal

cannot say

$4.7 + 2.85$ equals:

7.55

6.55

7.12

4.98

$9.1 - 3.62$ equals:

5.48

6.48

5.52

6.52

$3 + 0.45$ equals:

0.75

3.45

7.5

3.045

$5 - 1.35$ equals:

3.65

4.65

3.75

4.35

$0.8 + 0.9$ equals:

0.17

1.7

1.07

0.98

An estimate of $4.7 + 2.85$ is:

8

80

0.8

7.55

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Decimal	O‘nli kasr	Десятичная дробь
Decimal point	O‘nli nuqta (vergul)	Десятичная запятая
Tenth	O‘ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Thousandth	Mingdan bir	Тысячная
To order	Tartiblash	Упорядочить
Trailing zero	Oxirgi nol	Конечный ноль
To line up	Ustma-ust qo‘yish	Выровнять

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To compare decimals you:

count the digits

compare column by column

add them

round them

0.65 compared with 0.7 is:

larger

smaller

equal

undetermined

To add decimals you line up:

the last digits

the decimal points

the first digits

nothing

9.1 written to two decimal places is:

9.01

9.10

9.1

0.91

$12.6 + 0.47$ equals:

13.07

12.13

17.6

13.7

An estimate mainly checks:

the digits

the decimal point

the signs

the units

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Larger: 0.7 or 0.65

ANSWER 0.7

2. Larger: 0.09 or 0.1

ANSWER 0.1

3. Order 3.4 , 3.25 , 3.5

ANSWER $3.25 < 3.4 < 3.5$

4. $4.7 + 2.85$

ANSWER 7.55

5. $12.6 + 0.47$

ANSWER 13.07

6. $3 + 0.45$

ANSWER 3.45

7. $0.8 + 0.9$

ANSWER 1.7

Medium · the standard question

7 PROBLEMS

1. $9.1 - 3.62$

ANSWER 5.48

2. $5 - 1.35$

ANSWER 3.65

3. $7.4 - 0.85$

ANSWER 6.55

4. $0.5 - 0.28$

ANSWER 0.22

5. Estimate $4.7 + 2.85$

ANSWER 8

6. Estimate $9.1 - 3.62$

ANSWER 5

7. Order $0.5, 0.45, 0.505$

ANSWER $0.45 < 0.5 < 0.505$

Hard · stretch and reason

7 PROBLEMS

1. $12.05 + 3.7 + 0.68$

ANSWER 16.43

2. $20 - 6.47$

ANSWER 13.53

3. $4.2 - 1.35 + 0.7$

ANSWER 3.55

4. Order 1.1 , 1.01 , 1.011 , 1.101

ANSWER $1.01 < 1.011 < 1.1 < 1.101$

5. The number halfway between 0.6 and 0.7

ANSWER 0.65

6. A length of 2.4 m less 85 cm

ANSWER 1.55 m

7. Why write 9.1 as 9.10 before subtracting?

ANSWER The hundredths column needs a digit to borrow from

07 Homework**Homework — 5 tasks**

Fill every number with trailing zeros so all have the same length.

1. Order 2.3 , 2.25 , 2.4 and 2.05 .
2. Compute $6.8 + 3.45$ and $0.9 + 0.75$.
3. Compute $8.2 - 4.67$ and $6 - 2.48$.
4. Estimate each of the four answers above and compare with your exact values.
5. A plank 3.5 m long has 68 cm cut off. How much is left, in metres?

Multiplying and dividing decimals

A Cambridge insert: count the decimal places for a product, and shift both numbers for a quotient.

LESSON 78–79

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 4.3–4.4

LESSON CLOCK

25'

25'

25'

3

2

Multiplying	25 min
Dividing by a whole number	25 min
Dividing by a decimal	25 min
Checking	3 min
Homework	2 min

LEARNING OBJECTIVES

- Multiply decimals by counting the decimal places.
- Divide a decimal by a whole number.
- Divide by a decimal by shifting both numbers.
- Check every answer against an estimate.

TEXTBOOK

National	pp. 217–222
Cambridge	Stage 7, pp. 48–54
Workbook	Exercise 4.3

01

Explanation

Multiplying

Multiply as if there were no points, then put in as many decimal places as the two numbers had between them.

PRODUCT	WHOLE -NUMBER VERSION	PLACES	ANSWER
$0.3 \cdot 0.4$	$3 \cdot 4 = 12$	2	0.12
$1.2 \cdot 5$	$12 \cdot 5 = 60$	1	6.0
$2.5 \cdot 0.4$	$25 \cdot 4 = 100$	2	1.00
$6.9 \cdot 5.2$	$69 \cdot 52 = 3588$	2	35.88

$0.3 \cdot 0.4$ IS 0.12 , NOT 1.2

Two decimal places go in because the two numbers had one each. Multiplying two numbers below 1 gives an answer smaller than either of them.

Dividing by a whole number

Divide as usual and keep the decimal point above where it was.

QUOTIENT	WORKING	ANSWER
$7.2 \div 4$	$72 \div 4 = 18$	1.8
$0.35 \div 5$	$35 \div 5 = 7$	0.07
$9 \div 4$	$9.00 \div 4$	2.25
$1.2 \div 8$	$1.200 \div 8$	0.15

ZEROS MAY BE ADDED AFTER THE POINT

$9 \div 4$ does not stop at 2 remainder 1: writing 9.00 lets the division finish exactly, at 2.25.

Dividing by a decimal

Shift the point in **both** numbers until the divisor is a whole number. The quotient is unchanged, because both have been multiplied by the same power of ten.

QUOTIENT	SHIFTED	ANSWER
$6 \div 0.2$	$60 \div 2$	30
$4.8 \div 0.6$	$48 \div 6$	8
$1.44 \div 0.12$	$144 \div 12$	12
$0.9 \div 0.05$	$90 \div 5$	18

SHIFT BOTH, BY THE SAME AMOUNT

$6 \div 0.2$ becomes $60 \div 2$, not $6 \div 2$. Shifting only the divisor changes the answer by a factor of ten.

Checking

CALCULATION	ESTIMATE	EXACT
$6.9 \cdot 5.2$	$7 \cdot 5 = 35$	35.88
$0.3 \cdot 0.4$	less than 0.4	0.12
$6 \div 0.2$	much more than 6	30
$7.2 \div 4$	about 2	1.8

TWO QUESTIONS BEFORE EVERY ANSWER

Should it be bigger or smaller than what I started with, and roughly how big? A decimal point in the wrong place fails both tests immediately.

02 Worked examples

EXAMPLE 1 Compute $6.9 \cdot 5.2$.

- 1 Ignore the points: $69 \cdot 52 = 3\,588$.
- 2 One place plus one place makes two.
- 3 $= 35.88$
Estimate $7 \cdot 5 = 35$ ✓

ANSWER

35.88

EXAMPLE 2 Compute $4.8 \div 0.6$.

- ① Shift both one place: $48 \div 6$.

- ② $= 8$

- ③ Sense check: the divisor is under 1, so the answer beats 4.8 ✓

ANSWER

8

EXAMPLE 3 Compute $9 \div 4$.

- ① Write 9 as 9.00.

- ② $9.00 \div 4$

- ③ $= 2.25$

ANSWER

2.25

03 Interactive model

Ask for 0.3×0.4 before any rule; the popular answer of 1.2 is worth writing on the board and then testing against “less than a half of 0.4”.

Places and shifts

$0.3 \cdot 0.4$ equals:

1.2

0.12

0.012

12

$1.2 \cdot 5$ equals:

0.6

6

60

6.5

$6.9 \cdot 5.2$ equals:

35.88

3.588

358.8

34.88

$7.2 \div 4$ equals:

1.8

18

0.18

2.8

$9 \div 4$ equals:

2.2

2.25

2.5

2 r 1

$6 \div 0.2$ equals:

3

12

30

1.2

$1.44 \div 0.12$ equals:

1.2

12

120

0.12

Multiplying two numbers below 1 gives:

a larger answer

a smaller answer than both

the same

a whole number

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply	Ko'paytirish	Умножать
To divide	Bo'lish	Делить
Decimal place	O'nli xona	Десятичный знак
Product	Ko'paytma	Произведение
Quotient	Bo'linma	Частное
To shift	Siljitmoq	Сдвигать
Whole-number divisor	Butun bo'luvchi	Целый делитель
Estimate	Baholash	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The number of decimal places in a product is:

the larger of the two

the sum of the two

always two

the smaller

$2.5 \cdot 0.4$ equals:

1

10

0.1

100

$0.35 \div 5$ equals:

0.7

0.07

7

0.007

To divide by 0.6 you first:

shift the dividend only

shift both one place

shift the divisor only

do nothing

$0.9 \div 0.05$ equals:

1.8

18

180

0.18

The best first check on a decimal answer is:

the digits

the size and the point

the sign

the units

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $0.3 \cdot 0.4$

ANSWER 0.12

2. $1.2 \cdot 5$

ANSWER 6

3. $2.5 \cdot 0.4$

ANSWER 1

4. $7.2 \div 4$

ANSWER 1.8

5. $0.35 \div 5$

ANSWER 0.07

6. $6 \div 0.2$

ANSWER 30

7. $4.8 \div 0.6$

ANSWER 8 **Medium · the standard question**

7 PROBLEMS

1. $6.9 \cdot 5.2$

ANSWER 35.88

2. $9 \div 4$

ANSWER 2.25

3. $1.2 \div 8$

ANSWER 0.15

4. $1.44 \div 0.12$

ANSWER 12

5. $0.9 \div 0.05$

ANSWER 18

6. $0.25 \cdot 0.8$

ANSWER 0.2

7. $3.6 \div 0.4$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $0.125 \cdot 0.8$

ANSWER 0.1

2. $2.4 \cdot 1.5$

ANSWER 3.6

3. $7 \div 0.25$

ANSWER 28

4. $0.084 \div 0.7$

ANSWER 0.12

5. A pen costs 3.5 thousand sum: 12 pens

ANSWER 42 thousand

6. 1.6 kg shared into 0.2 kg bags

ANSWER 8 bags

7. Why does $6 \div 0.2$ give 30?

ANSWER Thirty two-tenths make six

07 Homework

Homework — 5 tasks

Estimate first, calculate second, and compare the two.

1. Compute $0.6 \cdot 0.7$, $3.4 \cdot 2.5$ and $1.25 \cdot 0.4$.
2. Compute $8.4 \div 6$ and $11 \div 8$.
3. Compute $9 \div 0.3$ and $2.4 \div 0.08$.
4. Estimate each answer above before finding it exactly.
5. A rope 7.2 m long is cut into 0.6 m pieces. How many pieces?

Recall — angles

Naming, measuring and estimating angles before the calculating begins.

LESSON 80

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 5 ENTRY

LESSON CLOCK

10'

12'

12'

6'

Naming	10 min
Measuring	12 min
Estimating	12 min
Naming with three letters	6 min

LEARNING OBJECTIVES

- Name angles as acute, right, obtuse, straight or reflex.
- Measure and draw an angle with a protractor.
- Estimate an angle before measuring it.
- Use the three-letter naming of an angle.

TEXTBOOK

National	pp. 223–226
Cambridge	Stage 7, pp. 55–58
Workbook	Exercise 5.1

01

Explanation

Naming

NAME	SIZE	EVERYDAY EXAMPLE
acute	between 0° and 90°	a slice of pizza
right	90°	the corner of a book
obtuse	between 90° and 180°	a laptop half open
straight	180°	a flat line
reflex	between 180° and 360°	the outside of a corner
full turn	360°	once round

A RIGHT ANGLE IS THE RULER FOR THE OTHERS

Acute is less than a corner, obtuse is more. Holding up a corner of paper is a perfectly good way to classify an angle before measuring it.

Measuring

STEP	WHAT TO DO
1	put the centre of the protractor on the vertex
2	lay the base line along one arm
3	read where the other arm crosses the scale
4	choose the scale that agrees with your estimate

TWO SCALES, ONE RIGHT ANSWER

A protractor reads 50° and 130° at the same place. Estimating first — acute or obtuse? — decides which to take, and it is why step 4 exists.

Estimating

LANDMARK	SIZE	RECOGNISED BY
a right angle	90°	a square corner
half of it	45°	a diagonal of a square
a third of it	30°	a thin wedge
two right angles	180°	a straight line
a little over a right angle	100°	just past square

ESTIMATE TO THE NEAREST LANDMARK FIRST

“A bit more than 90° ” is enough to reject a reading of 80° . The estimate is not the answer; it is the check on the answer.

Naming with three letters

$\angle ABC$ means the angle at B , between the arms BA and BC . The middle letter is always the vertex.

WRITTEN	VERTEX	ARMS
$\angle ABC$	B	BA and BC
$\angle XYZ$	Y	YX and YZ
$\angle B$	B	only if there is no ambiguity

THE MIDDLE LETTER IS THE VERTEX

$\angle ABC$ and $\angle CBA$ are the same angle; $\angle BAC$ is a different one. Reading the middle letter first is the habit to build.

02 Worked examples

EXAMPLE 1 Classify angles of 42° , 90° , 137° and 210° .

- 1 42° — acute.
- 2 90° — right.
- 3 137° — obtuse.
- 4 210° — reflex.

ANSWER

Acute, right, obtuse, reflex

EXAMPLE 2 A protractor reads either 50° or 130° . The angle looks clearly less than a corner. Which is it?

① Less than a right angle means acute.

② Acute means under 90° .

③ 50°

ANSWER

50°

EXAMPLE 3 In $\angle PQR$, which point is the vertex?

① The middle letter names the vertex.

② Q.
The arms are QP and QR .

ANSWER

Q

03 Interactive model

Have the class estimate ten angles on the board before any protractor comes out; measurement errors of a hundred degrees disappear once estimating is a habit.

Name that angle

42° is:

acute

right

obtuse

reflex

137° is:

acute

right

obtuse

reflex

210° is:

acute

obtuse

straight

reflex

180° is:

obtuse

straight

reflex

a full turn

A full turn is:

180°

270°

360°

90°

In $\angle ABC$ the vertex is:

A

B

C

any of them

$\angle ABC$ and $\angle CBA$ are:

the same angle

different angles

supplementary

complementary

A protractor showing 50 and 130 for an obtuse angle means:

50°

130°

either

neither

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Arm	Tomon	Сторона
Acute angle	O‘tkir burchak	Острый угол
Right angle	To‘g‘ri burchak	Прямой угол
Obtuse angle	O‘tmas burchak	Тупой угол
Reflex angle	Yoyiq burchakdan katta	Развёрнутый и больше
Protractor	Transportir	Транспортир

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An acute angle is:

under 90°

over 90°

exactly 90°

over 180°

An obtuse angle is:

under 90°

between 90° and 180°

over 180°

exactly 180°

A reflex angle is:

under 90°

between 90° and 180°

between 180° and 360°

exactly 360°

The protractor is placed with its centre on:

Estimating before measuring decides:

In $\angle XYZ$ the arms are:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify 42°

ANSWER Acute

2. Classify 90°

ANSWER Right

3. Classify 137°

ANSWER Obtuse

4. Classify 180°

ANSWER Straight

5. Classify 210°

ANSWER Reflex

6. A full turn

ANSWER 360°

7. The vertex of $\angle ABC$

ANSWER B

Medium · the standard question

7 PROBLEMS

1. Classify 89° and 91°

ANSWER Acute and obtuse

2. A protractor reads 50 or 130 for an acute angle

ANSWER 50°

3. Half a right angle

ANSWER 45°

4. A third of a right angle

ANSWER 30°

5. Two right angles

ANSWER 180°

6. Three right angles

ANSWER 270°

7. Are $\angle ABC$ and $\angle CBA$ the same?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. The reflex angle beside an angle of 130°

ANSWER 230°

2. The reflex angle beside a right angle

ANSWER 270°

3. An angle and its reflex partner add to

ANSWER 360°

4. Why does a protractor carry two scales?

ANSWER So it can be laid along either arm

5. The smallest reflex angle

ANSWER Just over 180°

6. Estimate the angle between the hands at 3 o'clock

ANSWER 90°

7. And at 6 o'clock

ANSWER 180°

07 Homework

Homework — 5 tasks

Estimate every angle before you measure it, and write both numbers down.

1. Classify angles of 37° , 90° , 154° and 265° .
2. Draw angles of 35° , 115° and 90° with a protractor.
3. Measure four angles from your textbook, estimating each one first.
4. Name the vertex and the arms of $\angle MNP$.
5. Find the reflex angle that goes with an angle of 145° .

Unknown angles at a point and on a line

Four facts about angles that meet — and the equations they turn into.

LESSON 81–83

3 × 40 MIN

MATEMATIKA 6, §16

S7 5.1–5.2

LESSON CLOCK

25'

30'

35'

25'

5

The four facts	25 min
One unknown	30 min
Equations with letters	35 min
Writing reasons	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the fact that angles on a straight line add to 180° .
- Use the fact that angles at a point add to 360° .
- Use vertically opposite angles.
- Write and solve an equation for an unknown angle.

TEXTBOOK

National	pp. 227–236
Cambridge	Stage 7, pp. 55–62
Workbook	Exercise 5.2

01

Explanation

The four facts

SITUATION	FACT	REASON TO QUOTE
two angles making a straight line	they add to 180°	angles on a line
angles filling a turn	they add to 360°	angles at a point
two lines crossing	opposite angles are equal	vertically opposite
two angles making a right angle	they add to 90°	complementary angles

EVERY ANSWER CARRIES A REASON

In geometry the number alone is half the answer. “ $x = 55^\circ$, angles on a line” is the whole one, and the reason is where the mark is.

One unknown

GIVEN	FACT	WORKING	ANSWER
125° on a line with x	180°	$180 - 125$	55°
140° , 90° and x at a point	360°	$360 - 230$	130°
x vertically opposite 68°	equal	—	68°
x and 37° make a right angle	90°	$90 - 37$	53°

CHOOSE THE FACT BY THE PICTURE, NOT BY THE NUMBERS

Angles that look as if they fill a turn add to 360° even if there are five of them; angles along a line add to 180° however many there are.

Equations with letters

SITUATION	EQUATION	SOLUTION	THE ANGLES
x and $2x$ on a line	$3x = 180$	$x = 60$	60° , 120°
x , $x + 20$ and $2x$ at a point with 90°	$4x + 110 = 360$	$x = 62.5$	62.5° , 82.5° , 125°
$3x$ and $x + 40$ vertically opposite	$3x = x + 40$	$x = 20$	60°
x , $2x$, $3x$ on a line	$6x = 180$	$x = 30$	30° , 60° , 90°

GEOMETRY BECOMES ALGEBRA, AND BACK AGAIN

Write the equation from the picture, solve it with the methods of lesson 16, then put the value back into every expression to find the actual angles.

Writing reasons

STATEMENT	REASON
$x + 125 = 180$	angles on a straight line
$x = 55^\circ$	solving
$y = 68^\circ$	vertically opposite angles
$a + b + c = 360^\circ$	angles at a point

“IT LOOKS LIKE IT” IS NOT A REASON

A diagram is a guide, not evidence: angles are rarely drawn to scale. The reason must be one of the four facts, named.

02 Worked examples

EXAMPLE 1 Two angles on a straight line are 125° and x . Find x .

1 $x + 125 = 180$
Angles on a line.

2 $x = 55^\circ$

ANSWER

55°

EXAMPLE 2 Angles of x and $2x$ lie on a straight line. Find both.

① $x + 2x = 180$

② $3x = 180$, so $x = 60$.

③ The angles are 60° and 120° .

Check: they add to 180° ✓

ANSWER

60° and 120°

EXAMPLE 3 Two crossing lines make angles of $3x$ and $x + 40$ vertically opposite. Find them.

① Vertically opposite angles are equal.

② $3x = x + 40$

③ $2x = 40$, so $x = 20$.

④ Each angle is 60° .

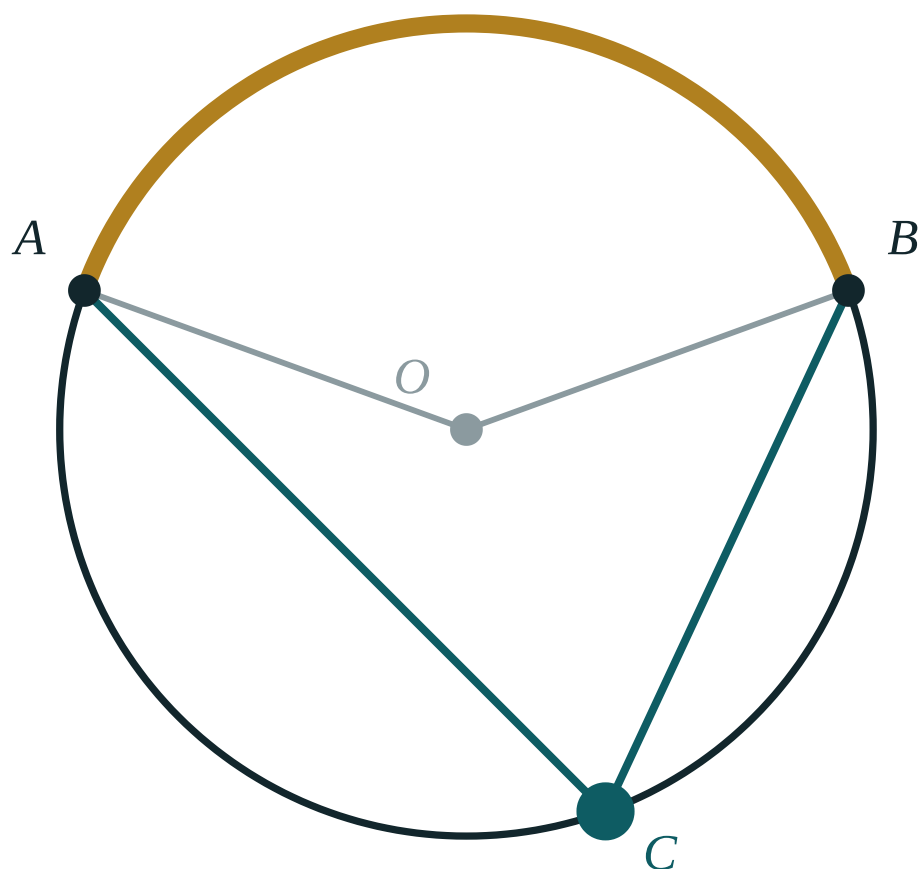
ANSWER

60°

03 Interactive model

Insist on a reason beside every line for a fortnight; the habit carries into Grade 7 proofs, where it becomes most of the mark.

Angles at a point add to 360°



central $\angle AOB$ 140°

inscribed $\angle ACB$ 70°

ratio **2**

arc AB 140°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Angles on a line	Yoyiq burchak	Смежные углы
Angles at a point	Nuqtadagi burchaklar	Углы вокруг точки
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Complementary	Yig'indisi 90°	Дополнительные до 90°
Supplementary	Yig'indisi 180°	Смежные
To justify	Asoslash	Обосновать
Unknown angle	Noma'lum burchak	Неизвестный угол
Reason	Sabab	Обоснование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Angles on a straight line add to:

Angles at a point add to:

Vertically opposite angles are:

125° on a line with x gives:

45°

55°

65°

235°

x and $2x$ on a line give x equal to:

45

60

90

120

A geometry answer needs:

the number

the number and a reason

a diagram

a measurement

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 125° on a line with x

ANSWER 55°

2. 90° on a line with x

ANSWER 90°

3. x vertically opposite 68°

ANSWER 68°

4. x and 37° make a right angle

ANSWER 53°

5. 140° , 90° and x at a point

ANSWER 130°

6. Angles on a line add to

ANSWER 180°

7. Angles at a point add to

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. x and $2x$ on a line

ANSWER 60° and 120°

2. x , $2x$, $3x$ on a line

ANSWER 30° , 60° , 90°

3. $3x$ and $x + 40$ vertically opposite

ANSWER $x = 20$, each 60°

4. x , x , x , x at a point

ANSWER 90° each

5. $2x$ and $x + 30$ on a line: find x

ANSWER $x = 50$

6. Four angles at a point, three of them 80° , 100° , 60°

ANSWER 120°

7. Two supplementary angles differ by 40°

ANSWER 110° and 70°

Hard · stretch and reason

7 PROBLEMS

1. x , $x + 20$, $2x$ and 90° at a point

ANSWER $x = 62.5^\circ$

2. Two complementary angles in the ratio 2 : 3

ANSWER 36° and 54°

3. Two supplementary angles in the ratio 4 : 5

ANSWER 80° and 100°

4. An angle is 3 times its supplement

ANSWER 135° and 45°

5. An angle equals its own complement

ANSWER 45°

6. An angle is 20° more than twice its complement

ANSWER $66\frac{2}{3}^\circ$

7. Why is “it looks equal” not a reason?

ANSWER Diagrams are rarely drawn to scale

07 Homework

Homework — 5 tasks

Write a reason beside every line of working.

- Find x if x and 118° lie on a straight line.
- Find x if x , $3x$ and 140° meet at a point.
- Find x if $5x$ and $2x + 60$ are vertically opposite.
- Two supplementary angles are in the ratio 3 : 7. Find both.

5. Find the angle that is twice its own complement.

Unknown angles in triangles and quadrilaterals

180° in a triangle, 360° in a quadrilateral, and the special figures that follow.

LESSON 84–86

3 × 40 MIN

MATEMATIKA 6, §17

S7 5.2

LESSON CLOCK

25'

30'

35'

25'

5

The angle sum of a triangle	25 min
Special triangles	30 min
Quadrilaterals	35 min
Figures made of triangles	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the angle sum of a triangle and of a quadrilateral.
- Use the base angles of an isosceles triangle.
- Find angles in figures made of several triangles.
- Write an equation for an unknown angle in a figure.

TEXTBOOK

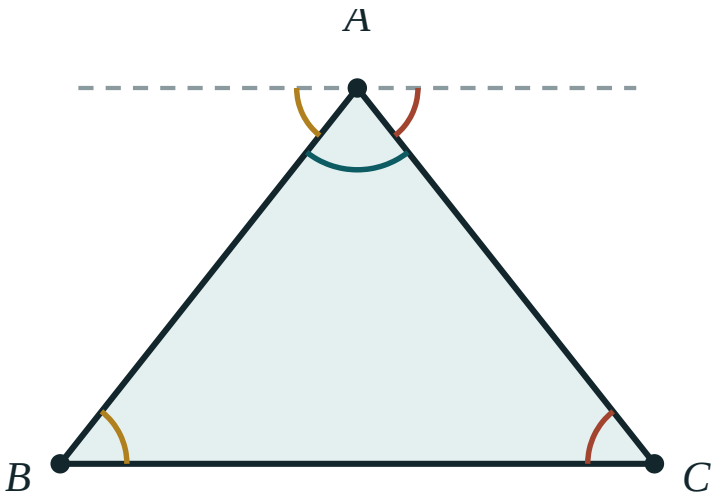
National	pp. 237–246
Cambridge	Stage 7, pp. 58–62
Workbook	Exercise 5.3

01

Explanation

The angle sum of a triangle

$$\angle A + \angle B + \angle C = 180^\circ$$



the three angles fill 180°

The three angles of a triangle

GIVEN	WORKING	THIRD ANGLE
$50^\circ, 60^\circ$	$180 - 110$	70°
$90^\circ, 35^\circ$	$180 - 125$	55°
$120^\circ, 25^\circ$	$180 - 145$	35°
angles $x, 2x, 3x$	$6x = 180$	$30^\circ, 60^\circ, 90^\circ$

TEAR THE CORNERS OFF AND CHECK

The three corners of any paper triangle lay flat along a straight line. It is not a proof, but it is convincing, and the proof waits until Grade 7.

Special triangles

KIND	WHAT IS EQUAL	ANGLES
equilateral	all three sides	$60^\circ, 60^\circ, 60^\circ$
isosceles	two sides	the two base angles are equal
right-angled	one angle is 90°	the other two add to 90°
right isosceles	two sides and a right angle	$90^\circ, 45^\circ, 45^\circ$

ISOSCELES TRIANGLE	WORKING	ANSWER
apex 40°	$(180 - 40) \div 2$	70° each
base angle 50°	$180 - 100$	80°

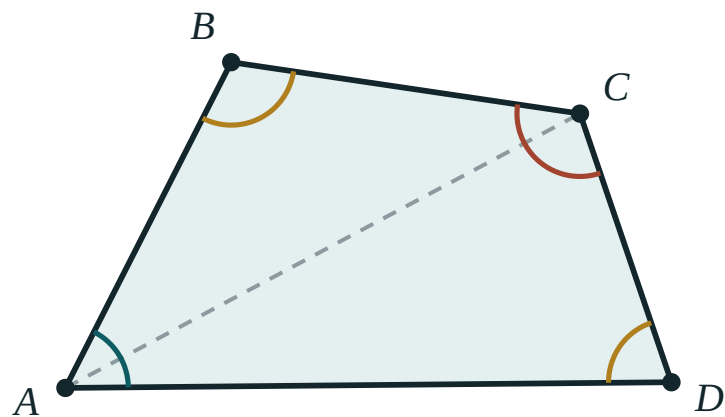
READ WHICH ANGLE IS GIVEN

An apex of 40° gives base angles of 70° ; a base angle of 40° gives an apex of 100° . The same number, two different triangles.

Quadrilaterals

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

A diagonal cuts any quadrilateral into two triangles, so its angles add to $2 \cdot 180^\circ$.



$$2 \times 180^\circ = 360^\circ$$

A quadrilateral split into two triangles

FIGURE	ANGLE FACTS
any quadrilateral	the four add to 360°
rectangle	four right angles
parallelogram	opposite angles equal; neighbours add to 180°
rhombus	as a parallelogram, and all sides equal
kite	one pair of opposite angles equal

TWO TRIANGLES, ONE QUADRILATERAL

That single picture explains the 360° , and it generalises: a pentagon splits into three triangles and has an angle sum of 540° .

Figures made of triangles

PROBLEM	ROUTE	ANSWER
a triangle with angles $x, x, 80^\circ$	$2x = 100$	50°
a quadrilateral with $90^\circ, 90^\circ, 130^\circ$	$360 - 310$	50°
a parallelogram with one angle 72°	neighbours add to 180°	108°
an exterior angle of a triangle, interior 115°	$180 - 115$	65°

WORK OUTWARDS FROM WHAT YOU KNOW

Fill in every angle you can, even ones the question did not ask for; the wanted angle usually appears after two or three of them are written on the diagram.

02 Worked examples

EXAMPLE 1 A triangle has angles 50° and 60° . Find the third.

1 $50 + 60 = 110$

2 $180 - 110 = 70$

3 70°

ANSWER

70°

EXAMPLE 2 An isosceles triangle has an apex angle of 40° . Find the base angles.

- ① The two base angles are equal.

- ② $180 - 40 = 140$ shared between them.

- ③ $140 \div 2 = 70^\circ$ each.

ANSWER

70° each

EXAMPLE 3 A quadrilateral has angles 90° , 90° and 130° . Find the fourth.

- ① The four add to 360° .

- ② $90 + 90 + 130 = 310$

- ③ $360 - 310 = 50^\circ$

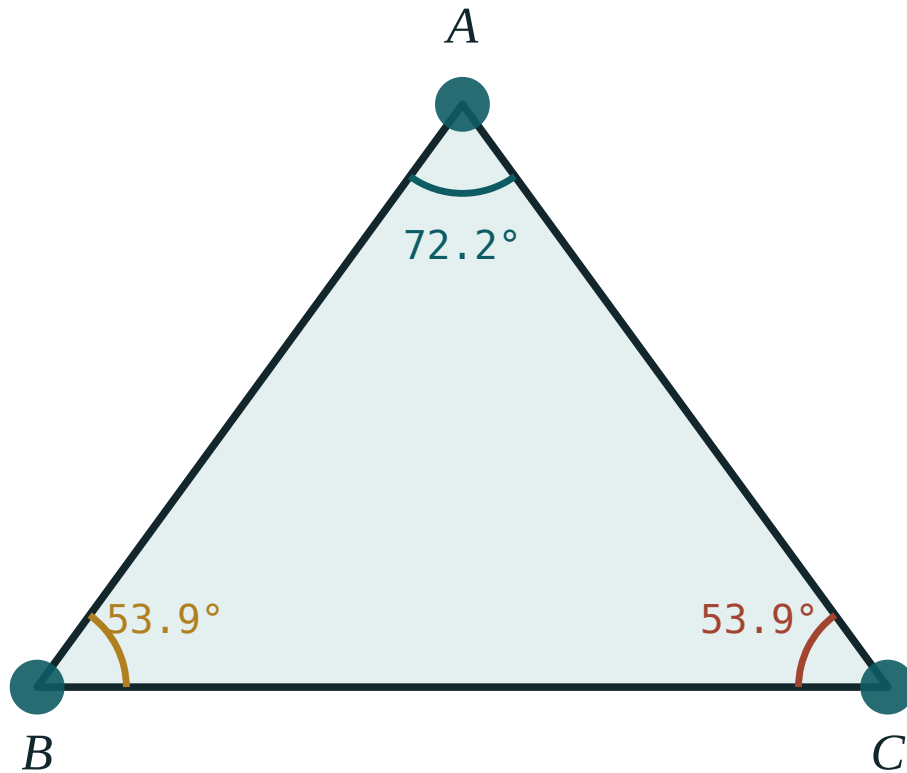
ANSWER

50°

03 Interactive model

Give a quadrilateral and ask the class to draw one diagonal; the 360° is then discovered rather than announced.

Drag a vertex — the sum stays at 180°



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side **BC**

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Triangle	Uchburchak	Треугольник
Quadrilateral	To'rtburchak	Четырёхугольник
Angle sum	Burchaklar yig'indisi	Сумма углов
Isosceles	Teng yonli	Равнобедренный
Equilateral	Teng tomonli	Равносторонний
Base angles	Asosdagi burchaklar	Углы при основании
Parallelogram	Parallelogramm	Параллелограмм
Diagonal	Diagonal	Диагональ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angles of a triangle add to:

The angles of a quadrilateral add to:

An equilateral triangle has angles of:

An isosceles triangle with apex 40° has base angles:

A parallelogram with one angle 72° has a neighbour of:

A pentagon's angles add to:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 50° and 60° : the third angle

ANSWER 70°

2. 90° and 35°

ANSWER 55°

3. 120° and 25°

ANSWER 35°

4. Each angle of an equilateral triangle

ANSWER 60°

5. Isosceles with apex 40° : the base angles

ANSWER 70°

6. Isosceles with base angle 50° : the apex

ANSWER 80°

7. A quadrilateral with 90° , 90° , 130°

ANSWER 50°

Medium · the standard question

7 PROBLEMS

1. A triangle with angles x , $2x$, $3x$

ANSWER $30^\circ, 60^\circ, 90^\circ$

2. A triangle with x , x , 80°

ANSWER 50° each

3. A parallelogram with one angle 72°

ANSWER 108°

4. A right isosceles triangle

ANSWER $90^\circ, 45^\circ, 45^\circ$

5. A quadrilateral with x , x , $2x$, $2x$

ANSWER $60^\circ, 60^\circ, 120^\circ, 120^\circ$

6. An exterior angle beside an interior angle of 115°

ANSWER 65°

7. The angle sum of a pentagon

ANSWER 540°

Hard · stretch and reason

7 PROBLEMS

1. A triangle with angles in the ratio $2 : 3 : 4$

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. A quadrilateral with angles in the ratio $1 : 2 : 3 : 4$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

3. An isosceles triangle with one angle 100° : the others

ANSWER 40° each

4. An isosceles triangle with one angle 40° : two possibilities

ANSWER $40^\circ, 70^\circ, 70^\circ$ or $40^\circ, 40^\circ, 100^\circ$

5. A kite with angles $110^\circ, 80^\circ, 80^\circ$: the fourth

ANSWER 90°

6. Why do a quadrilateral's angles add to 360° ?

ANSWER A diagonal splits it into two triangles

7. The angle sum of a hexagon

ANSWER 720°

07 Homework

Homework — 5 tasks

Fill in every angle you can find, not only the one asked for.

1. A triangle has angles 72° and 43° . Find the third.
2. An isosceles triangle has an apex angle of 34° . Find the base angles.
3. A quadrilateral has angles $85^\circ, 95^\circ$ and 110° . Find the fourth.
4. A triangle has angles in the ratio $3 : 4 : 5$. Find them.
5. A parallelogram has one angle of 115° . Find all four.

Control work 5 — angles

Angles on a line, at a point, in triangles and in quadrilaterals.

LESSON 87

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 5

S7 5 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Classify and measure angles accurately.
- Use the four angle facts with a reason each time.
- Find angles in triangles and quadrilaterals.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 223–246
Cambridge	Stage 7, pp. 55–62
Workbook	Control paper 5

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	Classify 37° , 90° , 128° and 245°	2	L80
2	x and 118° on a straight line: find x with a reason	3	L81–83
3	x , $3x$ and 140° at a point: find x	3	L81–83
4	$5x$ and $2x + 60$ vertically opposite: find the angle	3	L81–83
5	A triangle with 72° and 43° : find the third	3	L84–86
6	An isosceles triangle with apex 34° : find the base angles	3	L84–86
7	A quadrilateral with 85° , 95° and 110° : find the fourth	3	L84–86

THE ANSWERS

acute, right, obtuse, reflex; 62° ; 55° ; 100° ; 65° ; 73° each; 70° .

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
reflex called obtuse	245°	over 180° is reflex
no reason given	$x = 62^\circ$	“angles on a straight line”
angles at a point summed to 180°	$x + 3x + 140 = 180$	$x + 3x + 140 = 360$
vertically opposite set equal to 180°	$5x + 2x + 60 = 180$	$5x = 2x + 60$
angle sum taken as 360° for a triangle	$360 - 115$	$180 - 115$
apex used as a base angle	$34^\circ, 34^\circ, 112^\circ$	$34^\circ, 73^\circ, 73^\circ$
quadrilateral summed to 180°	a negative answer	360°

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	IT RETURNS IN
Q1	every diagram for the rest of the year
Q2–Q4	the circle, where angles at the centre fill 360°
Q5–Q6	Grade 7, where these facts are proved
Q7	the pie chart in Quarter IV

LOOKING FORWARD

The rest of Quarter II is the circle: its parts, and the length of the curve round it. The 360° of this paper is what makes a pie chart possible.

02 Worked examples

EXAMPLE 1 Model answer, Q3: x , $3x$ and 140° at a point.

① $x + 3x + 140 = 360$

Angles at a point.

② $4x = 220$

③ $x = 55^\circ$

Check: $55 + 165 + 140 = 360$ ✓

ANSWER

55°

EXAMPLE 2 Model answer, Q4: $5x$ and $2x + 60$ vertically opposite.

① Vertically opposite angles are equal.

② $5x = 2x + 60$

③ $3x = 60$, so $x = 20$.

④ Each angle is 100° .

ANSWER

100°

EXAMPLE 3 Model answer, Q6: isosceles with apex 34° .

① $180 - 34 = 146$ for the two base angles.

② $146 \div 2 = 73$.

③ 73° each.

Check: $34 + 73 + 73 = 180$ ✓

ANSWER

73° each

03 Interactive model

Mark Q2 for the reason alone; the class discovers that the number without a reason is worth less than half the marks.

The chapter in eight questions

245° is:

acute

obtuse

straight

reflex

x with 118° on a line:

62°

72°

242°

118°

Its reason:

angles at a point

angles on a straight line

vertically opposite

triangle sum

$$x + 3x + 140 = \dots$$

So x is:

$5x$ and $2x + 60$ vertically opposite give the angle:

A triangle with 72° and 43° :

A quadrilateral with 85° , 95° , 110° :

60°

70°

80°

90°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Acute angle	O‘tkir burchak	Острый угол
Obtuse angle	O‘tmas burchak	Тупой угол
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Angle sum	Burchaklar yig‘indisi	Сумма углов
Isosceles	Teng yonli	Равнобедренный
Reason	Asos	Обоснование
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

calculating

classifying

measuring

drawing

Q2 needs, besides the number:

a diagram

a reason

a measurement

nothing

Q3 sums to:

 90° 180° 360° 540°

Q4 sets the two expressions:

equal

to 180°

to 360°

to 90°

Q6 divides by:

2

3

4

nothing

Q7 sums to:

180°

270°

360°

540°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify 37°

ANSWER Acute

2. Classify 128°

ANSWER Obtuse

3. Classify 245°

ANSWER Reflex

4. x with 118° on a line

ANSWER 62°

5. x , $3x$, 140° at a point

ANSWER $x = 55^\circ$

6. A triangle with 72° and 43°

ANSWER 65°

7. A quadrilateral with 85° , 95° , 110°

ANSWER 70°

Medium · the standard question

7 PROBLEMS

1. $5x$ and $2x + 60$ vertically opposite

ANSWER 100°

2. Isosceles with apex 34°

ANSWER 73° each

3. Isosceles with base angle 34°

ANSWER Apex 112°

4. Two supplementary angles in the ratio 3 : 7

ANSWER 54° and 126°

5. An angle twice its complement

ANSWER 60°

6. A parallelogram with one angle 115°

ANSWER $115^\circ, 65^\circ, 115^\circ, 65^\circ$

7. The reflex angle beside 118°

ANSWER 242°

Hard · stretch and reason

7 PROBLEMS

1. A triangle with angles x , $x + 20$, $x + 40$

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. A quadrilateral with angles x , $2x$, $3x$, $4x$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

3. Four angles at a point, three equal and one 120°

ANSWER 80° each

4. An isosceles triangle with one angle of 90°

ANSWER $90^\circ, 45^\circ, 45^\circ$

5. A triangle cannot have two obtuse angles: why?

ANSWER They already exceed 180°

6. An exterior angle of a triangle beside 65°

ANSWER 115°

7. Why is a reason needed for every angle answer?

ANSWER The diagram is not evidence; the fact is

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its reason.

1. Rewrite in full every question on which you lost a mark.
2. Find x if x and 143° lie on a straight line, giving a reason.
3. Find x if $2x$, $4x$ and 120° meet at a point.
4. An isosceles triangle has an apex of 50° . Find the base angles.
5. A quadrilateral has angles 100° , 75° and 95° . Find the fourth.

Think — angles in design

Tiles, patterns and the 360° that has to be filled at every meeting point.

LESSON 88

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 5.3

LESSON CLOCK

10'

12'

12'

6'

The angles of a regular polygon	10 min
Which shapes tile	12 min
Why pentagons fail	12 min
Ornament	6 min

LEARNING OBJECTIVES

- Find the angles of a regular polygon.
- Decide which regular polygons tile the plane.
- Explain a tiling by the angles at a meeting point.
- Recognise the geometry in Uzbek ornament.

TEXTBOOK

National	pp. 247–249
Cambridge	Stage 7, pp. 62–65
Workbook	Exercise 5.4

01

Explanation

The angles of a regular polygon

interior angle = $\frac{180^\circ(n - 2)}{n}$

POLYGON	<i>N</i>	ANGLE SUM	EACH ANGLE
triangle	3	180°	60°
square	4	360°	90°
pentagon	5	540°	108°
hexagon	6	720°	120°
octagon	8	1080°	135°

THE ANGLE SUM COMES FROM TRIANGLES

An n -sided polygon splits into $n - 2$ triangles from one vertex, so its angles add to $180^\circ(n - 2)$. A regular one shares that equally.

Which shapes tile

Tiles meet without gaps only if the angles at each meeting point add to exactly 360° .

SHAPE	ANGLE	$360 \div \text{ANGLE}$	TILES?
triangle	60°	6	yes
square	90°	4	yes
pentagon	108°	3.33...	no
hexagon	120°	3	yes
octagon	135°	2.67...	no, alone

THE TEST IS A WHOLE-NUMBER DIVISION

Only three regular polygons tile the plane by themselves: the triangle, the square and the hexagon. Bees use the third of them.

Why pentagons fail

PENTAGONS AT A POINT	TOTAL	VERDICT
3	324°	a gap of 36°
4	432°	an overlap of 72°

Octagons alone fail too — but octagons and squares together work: $135 + 135 + 90 = 360$, which is the pattern on countless tiled floors.

MIXED TILINGS OPEN THE QUESTION UP AGAIN

Squares and triangles: $90 + 90 + 60 + 60 + 60 = 360$. Hexagons and triangles: $120 + 120 + 60 + 60 = 360$. Each combination is a small piece of arithmetic with a floor pattern as its answer.

Ornament

WHERE	SHAPES USED	ANGLE FACT BEHIND IT
a girih tile panel	stars and hexagons	angles meeting at 360°
a brick floor	rectangles	$90^\circ \cdot 4$
a honeycomb	hexagons	$120^\circ \cdot 3$
a tiled dome	triangles and squares	mixed tilings

THE ORNAMENT OF SAMARKAND IS ANGLE ARITHMETIC

The star patterns on the madrasahs are built from polygons whose angles fit exactly round a point. The craftsmen worked it out with compasses and a straightedge, five centuries before the formula was written the way it is above.

02 Worked examples

EXAMPLE 1 Find each interior angle of a regular hexagon.

- 1 Angle sum: $180 \cdot (6 - 2) = 720^\circ$.
- 2 Shared equally between six angles.
- 3 $720 \div 6 = 120^\circ$

ANSWER

120°

EXAMPLE 2 Do regular pentagons tile the plane?

- ① Each angle is 108° .
- ② $360 \div 108 = 3.33\dots$ — not a whole number.
- ③ Three leave a gap of 36° ; four overlap.
- ④ So no.

ANSWER

No

EXAMPLE 3 Show that octagons and squares tile together.

- ① An octagon angle is 135° , a square angle 90° .
- ② $135 + 135 + 90 = 360$
- ③ Two octagons and a square meet exactly at each point.

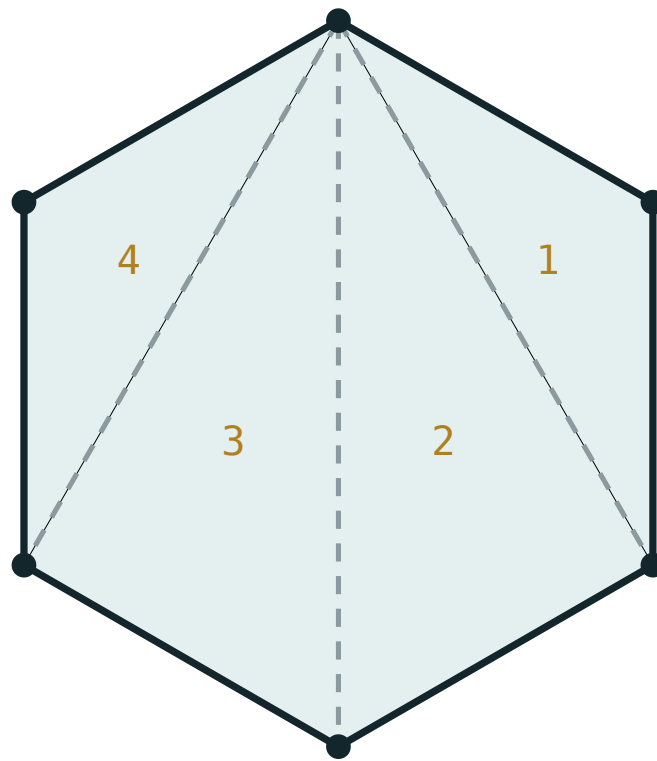
ANSWER

They tile

03 Interactive model

Cut out card pentagons and let the class try to tile a desk; the stubborn gap is a better memory than the arithmetic that predicts it.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**triangles **4**interior sum **720°**each angle **120°**exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Regular polygon	Muntazam ko'pburchak	Правильный многоугольник
Tiling	Qoplama	Замошение
Meeting point	Uchrashuv nuqtasi	Точка стыка
Interior angle	Ichki burchak	Внутренний угол
Hexagon	Olti burchak	Шестиугольник
Pentagon	Besh burchak	Пятиугольник
Ornament	Naqsh	Орнамент
Symmetry	Simmetriya	Симметрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angle sum of a pentagon is:

Each angle of a regular hexagon is:

Tiles meet without gaps when the angles add to:

Regular pentagons:

tile

do not tile

tile with triangles only

tile in threes

How many regular polygons tile alone?

1

2

3

4

Two octagons and a square meet at:

300°

330°

360°

390°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The angle sum of a quadrilateral

ANSWER 360°

2. The angle sum of a pentagon

ANSWER 540°

3. The angle sum of a hexagon

ANSWER 720°

4. Each angle of an equilateral triangle

ANSWER 60°

5. Each angle of a regular pentagon

ANSWER 108°

6. Each angle of a regular hexagon

ANSWER 120°

7. Each angle of a regular octagon

ANSWER 135°

Medium · the standard question

7 PROBLEMS

1. Do triangles tile?

ANSWER Yes — six at a point

2. Do squares tile?

ANSWER Yes — four at a point

3. Do pentagons tile?

ANSWER No

4. Do hexagons tile?

ANSWER Yes — three at a point

5. Do octagons tile alone?

ANSWER No

6. Octagons and squares together?

ANSWER Yes — $135 + 135 + 90$

7. The angle sum of a decagon

ANSWER 1440°

Hard · stretch and reason

7 PROBLEMS

1. Each angle of a regular decagon

ANSWER 144°

2. Do decagons tile alone?

ANSWER No — $360 \div 144 = 2.5$

3. Squares and triangles at a point: one arrangement

ANSWER $90 + 90 + 60 + 60 + 60 = 360$

4. Hexagons and triangles at a point

ANSWER $120 + 120 + 60 + 60 = 360$

5. A regular polygon with interior angle 150° : how many sides?

ANSWER 12

6. Why does the interior angle grow with n ?

ANSWER The angle sum grows faster than the number of angles

7. The exterior angles of any polygon add to

ANSWER 360°

07 Homework

Homework — 5 tasks

Test every tiling claim by adding the angles at one meeting point.

- Find each interior angle of a regular pentagon, hexagon and octagon.
- Show by arithmetic which of them tile the plane alone.
- Find a combination of two different regular polygons that meet at 360° .
- Find the number of sides of a regular polygon whose interior angle is 156° .
- Photograph or draw one piece of Uzbek ornament and name the polygons in it.

Recall — the circle and the disc

The curve and the region it encloses are two different things, with two different names.

LESSON 89

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8.2 ENTRY

LESSON CLOCK

10'

12'

12'

6'

Circle and disc	10 min
The parts	12 min
Radius and diameter	12 min
Drawing one	6 min

LEARNING OBJECTIVES

- Distinguish a circle from a disc.
- Name the centre, radius, diameter, chord and arc.
- Draw a circle of a given radius with compasses.
- State the relation between radius and diameter.

TEXTBOOK

National	pp. 250–253
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.1

01

Explanation

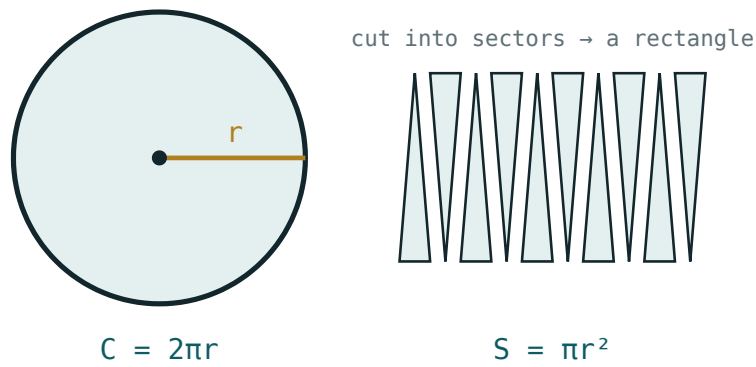
Circle and disc

WORD	MEANS	MEASURED BY
circle	the curve itself	a length — the circumference
disc	the curve and everything inside	an area

A CIRCLE IS A SET OF POINTS AT ONE DISTANCE FROM THE CENTRE

Every point of the curve is exactly r from the centre — no nearer, no further. That single sentence is the definition, and the compasses are its machine.

The parts



The centre, radius, diameter and chord

PART	WHAT IT IS
centre	the fixed point
radius	from the centre to the curve
diameter	right across, through the centre
chord	joining two points of the curve
arc	part of the curve
sector	a slice, between two radii
segment	the piece cut off by a chord

A DIAMETER IS A CHORD; A CHORD IS USUALLY NOT A DIAMETER

The diameter is the longest chord — the only one passing through the centre. Every other chord is shorter.

Radius and diameter

$$d = 2r \quad r = \frac{d}{2}$$

R	D
3 cm	6 cm
7 cm	14 cm
4.5 cm	9 cm
12 cm	24 cm

READ WHICH ONE THE QUESTION GIVES

A circle “of diameter 14 ” has radius 7 . Half the errors in the next three lessons come from using the wrong one of the two in a formula.

Drawing one

STEP	WHAT TO DO
1	open the compasses to the radius against a ruler
2	put the point on the centre and mark it
3	turn the compasses in one smooth movement
4	label the centre, and mark a radius and a diameter

SET THE COMPASSES TO THE RADIUS, NEVER THE DIAMETER

For a circle of diameter 10 cm the compasses open to 5 cm . The commonest practical error in this topic is drawing a circle twice the size wanted.

02 Worked examples**EXAMPLE 1** A circle has diameter *14 cm*. Find its radius.

$$① \quad r = \frac{d}{2}$$

$$② \quad = 7 \text{ cm}$$

ANSWER*7 cm***EXAMPLE 2** What is the difference between a circle and a disc?

① A circle is the curve alone.

② A disc is the curve together with the region inside it.

③ One is measured by a length, the other by an area.

ANSWER

Curve against region

EXAMPLE 3 Is every chord a diameter? Is every diameter a chord?

- ① A chord joins two points of the curve.
- ② A diameter does that and passes through the centre.
- ③ So every diameter is a chord; most chords are not diameters.

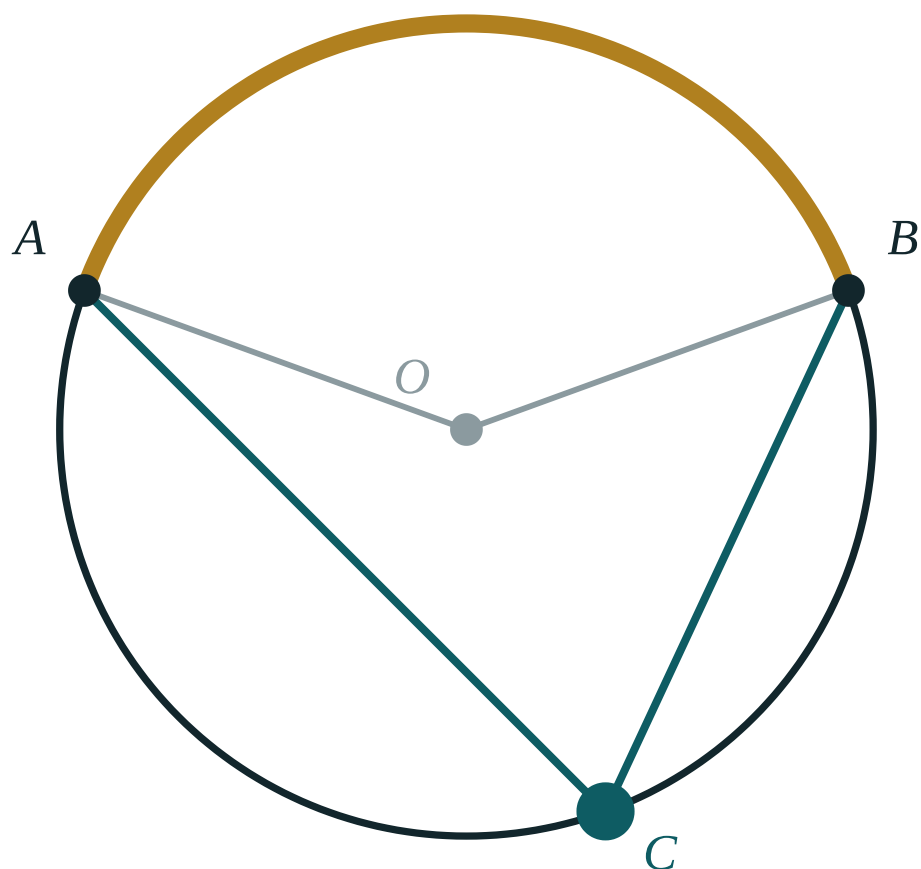
ANSWER

No; yes

03 **Interactive model**

Ask the class to draw a circle of diameter 8 cm; the ones that come out 16 cm across make the radius-diameter point better than any warning.

The parts of a circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Circle	Aylana	Окружность
Disc	Doira	Круг
Centre	Markaz	Центр
Radius	Radius	Радиус
Diameter	Diametr	Диаметр
Chord	Vatar	Хорда
Arc	Yoy	Дуга
Compasses	Sirkul	Циркуль

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A circle is:

the curve

the region inside

both

neither

A disc is measured by:

a length

an area

an angle

a radius only

The diameter is:

r

$2r$

$\frac{r}{2}$

πr

A circle of diameter *14* has radius:

28

7

14

3.5

The longest chord is:

a radius

the diameter

an arc

a sector

Compasses are set to:

the diameter

the radius

the circumference

any length

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The curve itself is called

ANSWER a circle

2. The region inside is called

ANSWER a disc

3. $r = 3$: the diameter

ANSWER 6 cm

4. $d = 14$: the radius

ANSWER 7 cm

5. $r = 4.5$: the diameter

ANSWER 9 cm

6. A line from the centre to the curve

ANSWER a radius

7. A line joining two points of the curve

ANSWER a chord

Medium · the standard question

7 PROBLEMS

1. The longest chord in a circle

ANSWER the diameter

2. Part of the curve

ANSWER an arc

3. A slice between two radii

ANSWER a sector

4. The piece cut off by a chord

ANSWER a segment

5. Is every diameter a chord?

ANSWER Yes

6. Is every chord a diameter?

ANSWER No

7. Compasses for a circle of diameter 10 cm open to

ANSWER 5 cm

Hard · stretch and reason

7 PROBLEMS

1. A circle of radius r : the longest chord

ANSWER $2r$

2. Two circles of radius 5 with centres 8 cm apart: do they cross?

ANSWER Yes — $8 < 10$

3. Two circles of radius 5 with centres 12 cm apart

ANSWER No — they are apart

4. A chord 6 cm from the centre of a circle of radius 5 cm

ANSWER Impossible — no such chord

5. How many diameters has a circle?

ANSWER Infinitely many

6. Where do all the diameters meet?

ANSWER At the centre

7. Why is a circle drawn with compasses?

ANSWER They keep the distance from the centre fixed

07 Homework

Homework — 5 tasks

Set the compasses to the radius; halve the diameter first if that is what you are given.

1. Draw circles of radius 3 cm and of diameter 9 cm.
2. Mark and label a radius, a diameter, a chord and an arc on one of them.
3. Find the radius of circles with diameters 18 cm and 7 cm.
4. Find the diameter of circles with radii 6.5 cm and 11 cm.
5. Write one sentence explaining the difference between a circle and a disc.

The elements of a circle and of a disc

Arcs, sectors and segments — and the angle at the centre that measures them.

LESSON 90–91

2 × 40 MIN

MATEMATIKA 6, §18

S7 8.2

LESSON CLOCK

20'

25'

25'

8'

2

Arcs and central angles	20 min
Sectors	25 min
Segments and the rest	25 min
Filling the turn	8 min
Homework	2 min

LEARNING OBJECTIVES

- Name every part of a circle and of a disc.
- Measure an arc by the angle it subtends at the centre.
- Distinguish a sector from a segment.
- Use the fact that the angles at the centre fill 360° .

TEXTBOOK

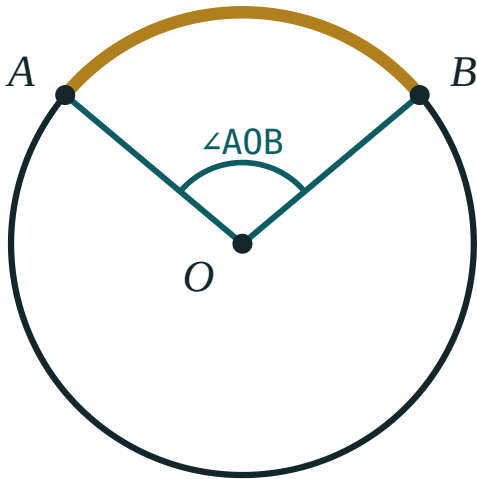
National	pp. 254–261
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.2

01

Explanation

Arcs and central angles

An **arc** is a piece of the curve. Its size is given by the **central angle** — the angle it makes at the centre.



a central angle equals the arc it cuts

An arc and the angle it makes at the centre

CENTRAL ANGLE	THE ARC IS	FRACTION OF THE CIRCLE
360°	the whole circle	1
180°	a semicircle	$\frac{1}{2}$
90°	a quarter	$\frac{1}{4}$
120°	a third	$\frac{1}{3}$
60°	a sixth	$\frac{1}{6}$

THE ANGLE IS THE MEASURE

Two arcs of the same circle are equal exactly when their central angles are equal. This is what makes a pie chart work: a share of the data becomes a share of 360° .

Sectors

A **sector** is a slice of the disc: the region between two radii and the arc between them.

CENTRAL ANGLE	NAME	FRACTION OF THE DISC
180°	half disc	$\frac{1}{2}$
90°	quadrant	$\frac{1}{4}$
45°	an eighth	$\frac{1}{8}$
72°	a fifth	$\frac{1}{5}$

A SECTOR IS A FRACTION OF THE WHOLE DISC

The fraction is $\frac{\alpha}{360}$. Everything about sectors — arc length, area, pie charts — comes from that one fraction.

Segments and the rest

PART	BOUNDED BY	NOT TO BE CONFUSED WITH
sector	two radii and an arc	a segment
segment	a chord and an arc	a sector
semicircle	a diameter and half the curve	a quadrant
tangent	touches the circle at one point	a chord

A SECTOR IS A SLICE OF PIZZA; A SEGMENT IS THE CRUST END

The sector reaches the centre; the segment does not. Naming them the wrong way round is the standard slip, and the picture fixes it.

Filling the turn

PROBLEM	WORKING	ANSWER
a disc split into 4 equal sectors	$360 \div 4$	90°
into 5 equal sectors	$360 \div 5$	72°
into 8 equal sectors	$360 \div 8$	45°
three sectors of 100° , 140° and x	$360 - 240$	120°

THIS IS THE PIE CHART, THREE MONTHS EARLY

Sharing 360° among categories is exactly the calculation of Quarter IV. The geometry is being built now so the statistics can use it later.

02 Worked examples**EXAMPLE 1** A disc is divided into 5 equal sectors. Find the central angle of each.

① The angles at the centre fill 360° .

② $360 \div 5$

③ $= 72^\circ$

ANSWER 72° **EXAMPLE 2** Three sectors of a disc have central angles 100° , 140° and x . Find x .

① $100 + 140 = 240$

② $360 - 240 = 120$

③ $x = 120^\circ$

ANSWER 120°

EXAMPLE 3 What fraction of a disc is a sector of 45° ?

① $\frac{45}{360}$

② Simplify by 45.

③ $\frac{1}{8}$

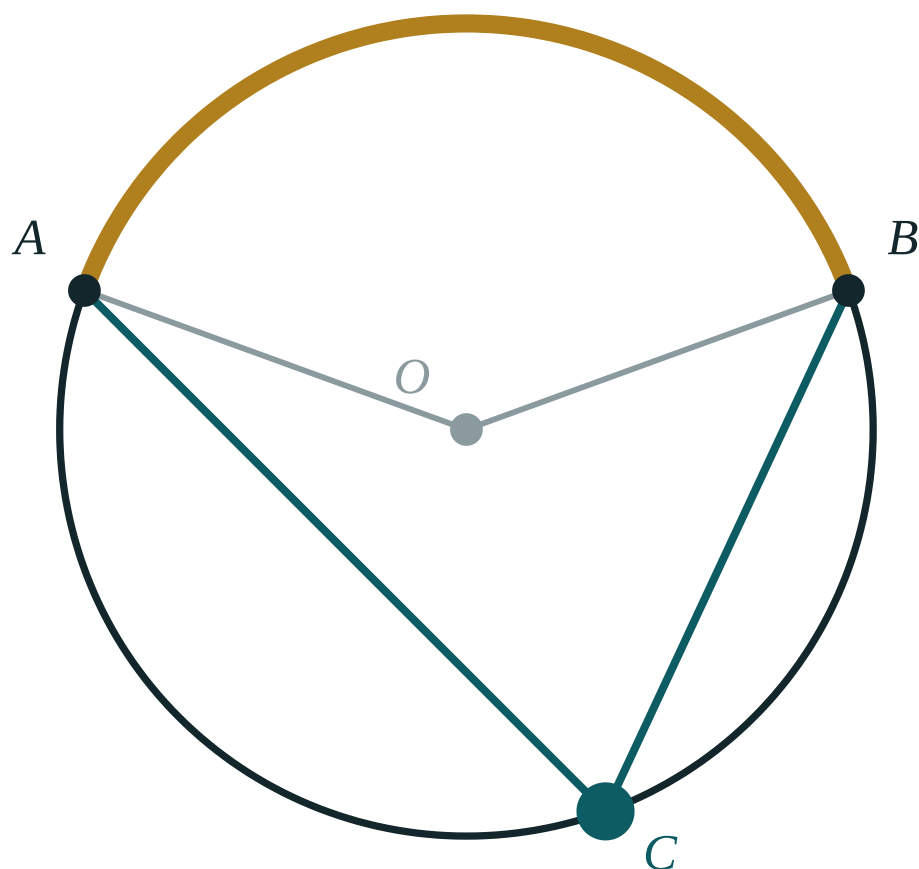
ANSWER

$$\frac{1}{8}$$

03 **Interactive model**

Cut a paper disc into sectors of different sizes and have the class name the angle of each; the fractions of 360° are then physical objects on the desk.

A sector and its central angle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Central angle	Markaziy burchak	Центральный угол
To subtend	Qarshi turmoq	Стягивать
Semicircle	Yarim aylana	Полуокружность
Quadrant	Chorak doira	Четверть круга
Tangent	Urinma	Касательная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An arc is measured by:

its chord

its central angle

the radius

the diameter

A sector is bounded by:

a chord and an arc

two radii and an arc

two chords

a tangent

A segment is bounded by:

a chord and an arc

two radii and an arc

two diameters

a tangent and a radius

A 90° sector is:

a semicircle

a quadrant

a segment

a third

Five equal sectors have central angles of:

60°

72°

90°

45°

A 45° sector is what fraction of the disc?

$\frac{1}{4}$

$\frac{1}{6}$

$\frac{1}{8}$

$\frac{1}{45}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A 180° arc is

ANSWER a semicircle

2. A 90° sector is

ANSWER a quadrant

3. A disc in 4 equal sectors: each angle

ANSWER 90°

4. A disc in 5 equal sectors

ANSWER 72°

5. A disc in 8 equal sectors

ANSWER 45°

6. A sector is bounded by

ANSWER two radii and an arc

7. A segment is bounded by

ANSWER a chord and an arc

Medium · the standard question

7 PROBLEMS

- Sectors of 100° , 140° and x

ANSWER $x = 120^\circ$

- A 120° sector as a fraction

ANSWER $\frac{1}{3}$

- A 45° sector as a fraction

ANSWER $\frac{1}{8}$

- A 72° sector as a fraction

ANSWER $\frac{1}{5}$

- A disc in 6 equal sectors

ANSWER 60°

- A disc in 12 equal sectors

ANSWER 30°

- A tangent touches the circle at

ANSWER one point

Hard · stretch and reason

7 PROBLEMS

1. A sector that is $\frac{3}{8}$ of a disc: its angle

ANSWER 135°

2. A sector of 216° as a fraction

ANSWER $\frac{3}{5}$

3. Four sectors in the ratio $1 : 2 : 3 : 4$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

4. Two sectors of 150° each and one more: the third

ANSWER 60°

5. A clock face: the angle between consecutive hour marks

ANSWER 30°

6. The angle swept by the minute hand in 20 minutes

ANSWER 120°

7. Why does a pie chart need 360° ?

ANSWER The whole data set fills one complete turn

07 Homework

Homework — 5 tasks

Every sector question is a fraction of 360° .

1. Draw a circle and mark on it a sector, a segment, a chord and an arc.
2. A disc is divided into 9 equal sectors. Find each central angle.
3. Sectors of 75° , 125° and x fill a disc. Find x .
4. What fraction of a disc is a sector of 108° ?

5. Find the angle of a sector that is $\frac{5}{12}$ of a disc.

The circumference of a circle

$C = \pi d$ — measured with thread, then trusted as a formula.

LESSON 92–94

3 × 40 MIN

MATEMATIKA 6, §19

S7 8.2

LESSON CLOCK

25'

30'

35'

25'

5

The measurement	25 min
The formula	30 min
Backwards	35 min
Wheels and edges	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the ratio $C \div d$ by measurement.
- Use $C = \pi d$ and $C = 2\pi r$.
- Find the radius or diameter from the circumference.
- Solve practical problems about wheels and circular edges.

TEXTBOOK

National	pp. 262–272
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.3

01

Explanation

The measurement

Wrap thread round several round objects and divide the length by the diameter.

OBJECT	D	C	$C \div D$
a coin	2.4 cm	7.5 cm	3.13
a glass	7 cm	22 cm	3.14
a plate	21 cm	66 cm	3.14
a bucket	28 cm	88 cm	3.14

THE SAME NUMBER, EVERY TIME

Whatever the size of the circle, the ratio lands near 3.14. That constant is π , and the small differences in the table are measuring error, not differences between the circles.

The formula

$$C = \pi d = 2\pi r$$

R	D	C WITH $\pi = 3.14$	WITH $\pi = \frac{22}{7}$
5	10	31.4	31.43
7	14	43.96	44
10	20	62.8	62.86
21	42	131.88	132

CHOOSE THE APPROXIMATION TO SUIT THE NUMBERS

$\frac{22}{7}$ is exact-looking whenever the radius is a multiple of 7; 3.14 suits everything else. Both are approximations to the same constant.

Backwards

GIVEN	WORKING	ANSWER
$C = 31.4$	$d = C \div \pi$	$d = 10$
$C = 44$	$r = C \div 2\pi$	$r = 7$
$C = 62.8$	$r = C \div 2\pi$	$r = 10$
$C = 132$	$d = C \div \pi$	$d = 42$

DIVIDING BY π GIVES THE DIAMETER, NOT THE RADIUS

$C \div \pi = d$, and the radius is half of that. Stopping one step early is the standard error in the reverse question.

Wheels and edges

PROBLEM	WORKING	ANSWER
a wheel of diameter 70 cm: one turn	$\frac{22}{7} \cdot 70$	220 cm
how many turns in 220 m	$22\ 000 \div 220$	100
a round pond of radius 14 m: the fence	$2 \cdot \frac{22}{7} \cdot 14$	88 m
a running track of radius 35 m: one lap	$2 \cdot \frac{22}{7} \cdot 35$	220 m
a bicycle wheel of radius 35 cm in 50 turns	$50 \cdot 220$	110 m

KEEP THE UNITS IN ONE SYSTEM

Turning 220 m into 22 000 cm before dividing avoids the factor-of-a-hundred error that this kind of problem invites.

02 Worked examples

EXAMPLE 1 A circle has radius 7 cm. Find its circumference, taking $\pi = \frac{22}{7}$.

① $C = 2\pi r$

② $= 2 \cdot \frac{22}{7} \cdot 7$

③ $= 44\text{ cm}$

ANSWER

44 cm

EXAMPLE 2 A circle has circumference 62.8 cm . Find its radius, taking $\pi = 3.14$.

① $C = 2\pi r$

② $62.8 = 6.28r$

③ $r = 10\text{ cm}$

ANSWER

10 cm

EXAMPLE 3 A wheel of diameter 70 cm rolls 220 m . How many turns?

① One turn: $\frac{22}{7} \cdot 70 = 220\text{ cm}$.

② 220 m is $22\,000\text{ cm}$.
Same units.

③ $22\,000 \div 220 = 100$

ANSWER

100 turns

03 Interactive model

Do the thread measurement in groups and tabulate every ratio on the board; π arrives as a discovery rather than a fact to be believed.

Circumference, forwards and backwards

$C \div d$ equals:

With $r = 7$ and $\pi = \underline{22} \ 7$, C is:

With $d = 10$ and $\pi = 3.14$, C is:

$C = 62.8$ gives r equal to:

5

10

20

31.4

$C = 44$ gives r equal to:

7

14

22

3.5

A wheel of diameter 70 cm covers per turn:

70 cm

140 cm

220 cm

440 cm

In 220 m it makes:

10 turns

100 turns

1000 turns

22 turns

$C \div \pi$ gives:

- the radius
- the diameter
- the area
- π

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Circumference	Aylana uzunligi	Длина окружности
Pi	Pi soni	Число пи
Diameter	Diametr	Диаметр
Radius	Radius	Радиус
Ratio	Nisbat	Отношение
Revolution	Aylanish	Оборот
Approximation	Taqribiy qiymat	Приближение
Formula	Formula	Формула

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π is the ratio of:

C to r

C to d

d to r

r to C

C equals:

πr

πd

πr^2

$2\pi d$

A circle of radius 5 has circumference about:

15.7

31.4

78.5

10

To find d from C you:

multiply by π

divide by π

divide by 2π

square it

A pond of radius 14 m needs a fence of about:

44 m

88 m

176 m

616 m

22 π is best used when the radius is a multiple of:

2

3

7

10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$: the circumference ($\pi = 3.14$)

ANSWER 31.4

2. $r = 7$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 44

3. $d = 20$: the circumference ($\pi = 3.14$)

ANSWER 62.8

4. $d = 42$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 132

5. $C \div d$

ANSWER π

6. $r = 10$: the circumference

ANSWER 62.8

7. $r = 21$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 132

Medium · the standard question

7 PROBLEMS

1. $C = 31.4$: the diameter

ANSWER 10

2. $C = 44$: the radius

ANSWER 7

3. $C = 62.8$: the radius

ANSWER 10

4. $C = 132$: the diameter

ANSWER 42

5. A wheel of diameter 70 cm: one turn

ANSWER 220 cm

6. Turns in 220 m

ANSWER 100

7. A pond of radius 14 m: the fence

ANSWER 88 m

Hard · stretch and reason

7 PROBLEMS

1. A track of radius 35 m: one lap

ANSWER 220 m

2. Laps in 1 100 m

ANSWER 5

3. A bicycle wheel of radius 35 cm in 50 turns

ANSWER 110 m

4. A circle whose circumference equals its diameter times 3.14 has radius 9: its C

ANSWER 56.52

5. Doubling the radius does what to the circumference?

ANSWER Doubles it

6. A circular table of circumference 220 cm: its diameter

ANSWER 70 cm

7. Why is $C \div d$ the same for every circle?

ANSWER All circles are the same shape, scaled

07 Homework

Homework — 5 tasks

Say which value of π you are using, and why.

1. Find the circumference of circles with $r = 3.5$ cm and $d = 28$ cm.
2. Find the radius of a circle with circumference 88 cm.
3. A wheel of diameter 56 cm makes 100 turns. How far does it go?
4. A circular flower bed has radius 7 m. Find the length of edging needed.
5. Measure three round objects at home and tabulate $C \div d$ for each.

The perimeter of a semicircle and of a quarter circle

Half the curve is not the whole perimeter — the straight edges have to be added.

LESSON 95–96

2 × 40 MIN

MATEMATIKA 6, §20

S7 15

LESSON CLOCK

20'

25'

25'

8'

2

Arc length	20 min
The semicircle	25 min
The quadrant	25 min
Compound shapes	8 min
Homework	2 min

LEARNING OBJECTIVES

- Find the length of an arc as a fraction of the circumference.
- Find the perimeter of a semicircle and of a quadrant.
- Find the perimeter of a shape made of arcs and straight edges.
- Distinguish arc length from perimeter.

TEXTBOOK

National	pp. 273–280
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.1

01

Explanation

Arc length

arc = $\frac{\alpha}{360} \times C$

CENTRAL ANGLE	FRACTION	ARC OF A CIRCLE WITH $C = 44$
180°	$\frac{1}{2}$	22
90°	$\frac{1}{4}$	11
120°	$\frac{1}{3}$	14.67
45°	$\frac{1}{8}$	5.5

EVERYTHING IS A FRACTION OF THE WHOLE CIRCLE

The fraction is the central angle over 360. Nothing else about arcs is needed at this stage.

The semicircle

The perimeter of a half disc is the curved half **plus the diameter**.

$P = \pi r + 2r$

R	CURVE πR	DIAMETER	PERIMETER
7	22	14	36
14	44	28	72
10	31.4	20	51.4

HALF THE CIRCUMFERENCE IS NOT THE PERIMETER

A half disc is bounded by a curve *and* a straight edge. Leaving out the diameter is the commonest error in the whole topic — and the shape it describes would be open.

The quadrant

A quarter disc is bounded by a quarter of the curve and **two radii**.

$P = \frac{1}{2}\pi r + 2r$

R	CURVE	TWO RADII	PERIMETER
7	11	14	25
14	22	28	50
10	15.7	20	35.7

COUNT THE STRAIGHT EDGES FROM THE PICTURE

A semicircle has one; a quadrant has two. Drawing the shape and tracing round it with a finger is a more reliable method than remembering formulae.

Compound shapes

SHAPE	PIECES OF THE BOUNDARY	PERIMETER WITH $R = 7$
a semicircle on a 14 by 10 rectangle	three sides plus the curve	$10 + 10 + 14 + 22 = 56$
a running track: two straights of 80 and two semicircular ends of radius 35	two straights plus a full circle	$160 + 220 = 380$
a quadrant cut from a square of side 7	two sides plus the curve	$7 + 7 + 11 = 25$

ONLY THE OUTSIDE COUNTS

Where the semicircle sits on the rectangle, the 14 cm edge is inside the shape and is not part of the perimeter — but the two ends of it are. Trace the boundary before adding anything.

02 Worked examples

EXAMPLE 1 Find the perimeter of a semicircle of radius 7 cm, taking $\pi = \frac{22}{7}$.

1

The curve: $\pi r = \frac{22}{7} \cdot 7 = 22$ cm.

2

The straight edge is the diameter: 14 cm.

3

$22 + 14 = 36$ cm.

ANSWER

36 cm

EXAMPLE 2 Find the perimeter of a quadrant of radius 14 cm.

① The circumference is $2 \cdot \frac{22}{7} \cdot 14 = 88$ cm.

② A quarter of it: 22 cm.

③ Two radii: 28 cm.

④ $22 + 28 = 50$ cm.

ANSWER*50 cm***EXAMPLE 3 A running track has two straights of 80 m and two semicircular ends of radius 35 m. Find one lap.**

① The two semicircles make one full circle.

② $2 \cdot \frac{22}{7} \cdot 35 = 220$ m.

③ The straights: 160 m.

④ $220 + 160 = 380$ m.

ANSWER*380 m***03 Interactive model**

Have the class trace the boundary of each shape with a finger and say aloud what they pass; the missing diameter stops being missed.

Curve plus straight edges

A semicircle of radius 7 has curve of length:

Its perimeter is:

A quadrant of radius 7 has curve of length:

Its perimeter is:

11

18

25

32

A semicircle of radius 14 has perimeter:

44

72

88

28

A 120° arc of a circle with $C = 44$ is:

11

14.67

22

5.5

Two semicircular ends of a track make:

half a circle

one full circle

two circles

a quadrant

The commonest error here is:

using π

forgetting the straight edges

using the radius

adding twice

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perimeter	Perimetr	Периметр
Arc length	Yoy uzunligi	Длина дуги
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Straight edge	To'g'ri tomon	Прямая сторона
Compound shape	Murakkab shakl	Составная фигура
Fraction of the circle	Aylananing qismi	Часть окружности
To add the edges	Tomonlarni qo'shish	Сложить стороны

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An arc of 90° is what fraction of the circle?

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{4}$$

$$\frac{1}{8}$$

The perimeter of a semicircle is:

$$\pi r$$

$$\pi r + 2r$$

$$2\pi r$$

$$\pi r + r$$

The perimeter of a quadrant is:

$$\frac{1}{2}\pi r$$

$$\frac{1}{2}\pi r + 2r$$

$$\frac{1}{4}\pi r + r$$

$$\pi r + 2r$$

A semicircle of radius 10 has perimeter about:

31.4

51.4

62.8

20

Two semicircular ends of equal radius make:

a semicircle

a full circle

a quadrant

nothing

A boundary edge inside a compound shape:

- counts
- does not count
- counts twice
- is halved

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. A semicircle of radius 7: the curve

ANSWER 22

2. Its perimeter

ANSWER 36

3. A semicircle of radius 14: the curve

ANSWER 44

4. Its perimeter

ANSWER 72

5. A quadrant of radius 7: the curve

ANSWER 11

6. Its perimeter

ANSWER 25

7. A 90° arc as a fraction of the circle

ANSWER $\frac{1}{4}$

Medium · the standard question

7 PROBLEMS

1. A quadrant of radius 14: its perimeter

ANSWER 50

2. A semicircle of radius 10 ($\pi = 3.14$)

ANSWER 51.4

3. A quadrant of radius 10

ANSWER 35.7

4. A 120° arc of a circle with $C = 44$

ANSWER 14.67

5. A 45° arc of the same circle

ANSWER 5.5

6. A track: straights 80 m, ends of radius 35 m

ANSWER 380 m

7. A quadrant cut from a square of side 7

ANSWER 25

Hard · stretch and reason

7 PROBLEMS

1. A semicircle on top of a 14 by 10 rectangle: the perimeter

ANSWER 56

2. A shape made of three quadrants of radius 7

ANSWER Curve 33 plus the straight edges

3. A semicircle of perimeter 36: its radius

ANSWER 7

4. A semicircular window of diameter 1.4 m: the frame length

ANSWER 3.6 m

5. A running track of two straights 100 m and ends of radius 28 m

ANSWER 376 m

6. Why add the diameter to a semicircle?

ANSWER The straight edge is part of the boundary

7. A quarter-circle path of radius 21 m: its curved length

ANSWER 33 m

07 Homework

Homework — 5 tasks

Trace the boundary with a finger and list every piece before adding.

1. Find the perimeter of a semicircle of radius 21 cm.
2. Find the perimeter of a quadrant of radius 21 cm.
3. Find the length of a 60° arc of a circle of radius 7 cm.
4. A semicircle sits on top of a rectangle 20 cm by 12 cm, on the 20 cm side. Find the whole perimeter.

5. A track has straights of 90 m and semicircular ends of radius 21 m. Find one lap.